

Software Engineering and Digital Arts Conservation (SEDA)

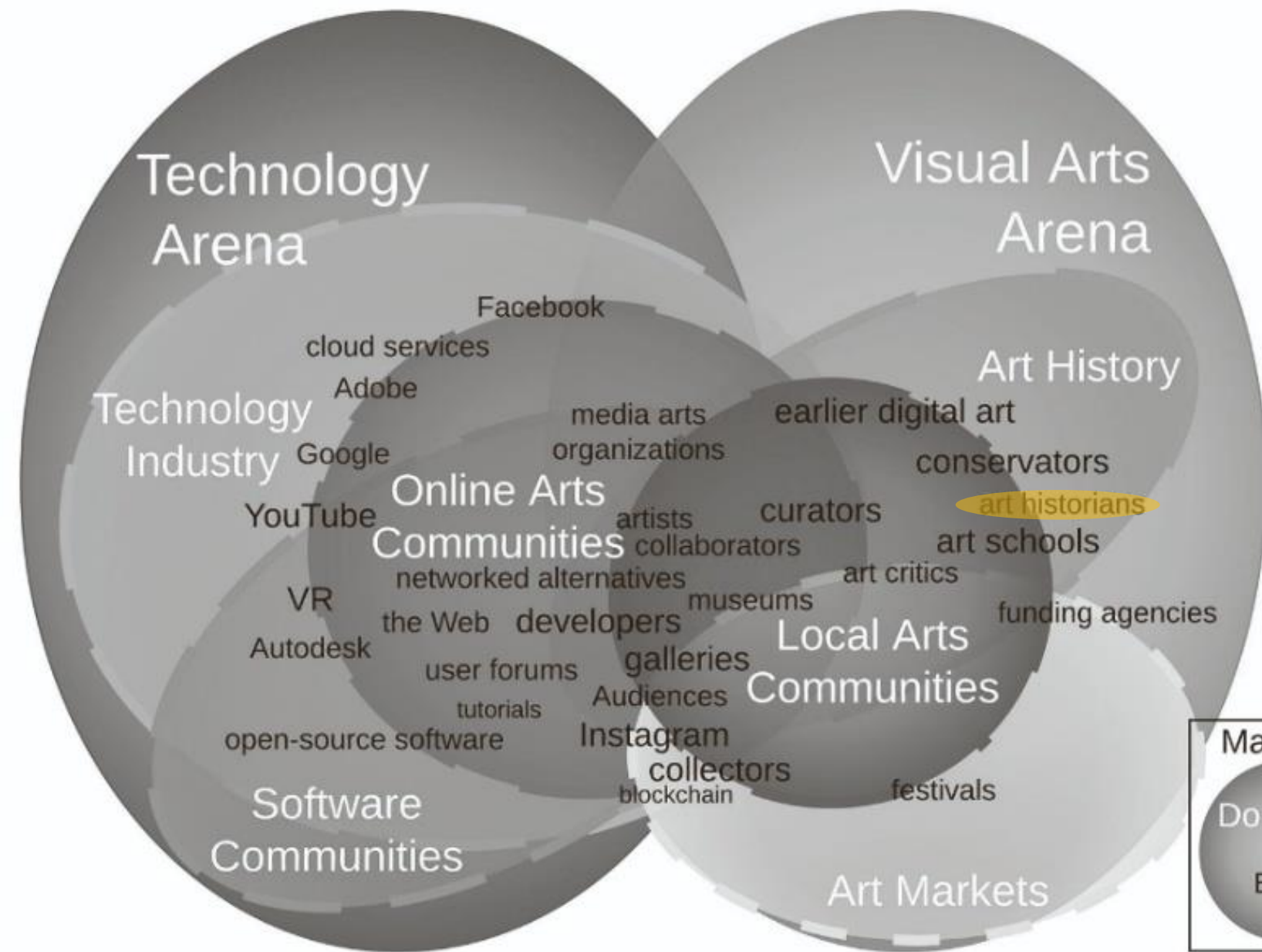
Art History

Inge Hinterwaldner (unibz)

Workshop, U Amsterdam, 24.3.2026

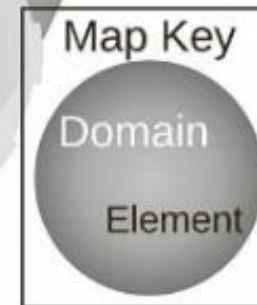


European Research Council
Established by the European Commission



What are art historians good for?

Model building (maybe?)



Social worlds and arenas map of artists and curators involved in the care of net-based art.

Source:

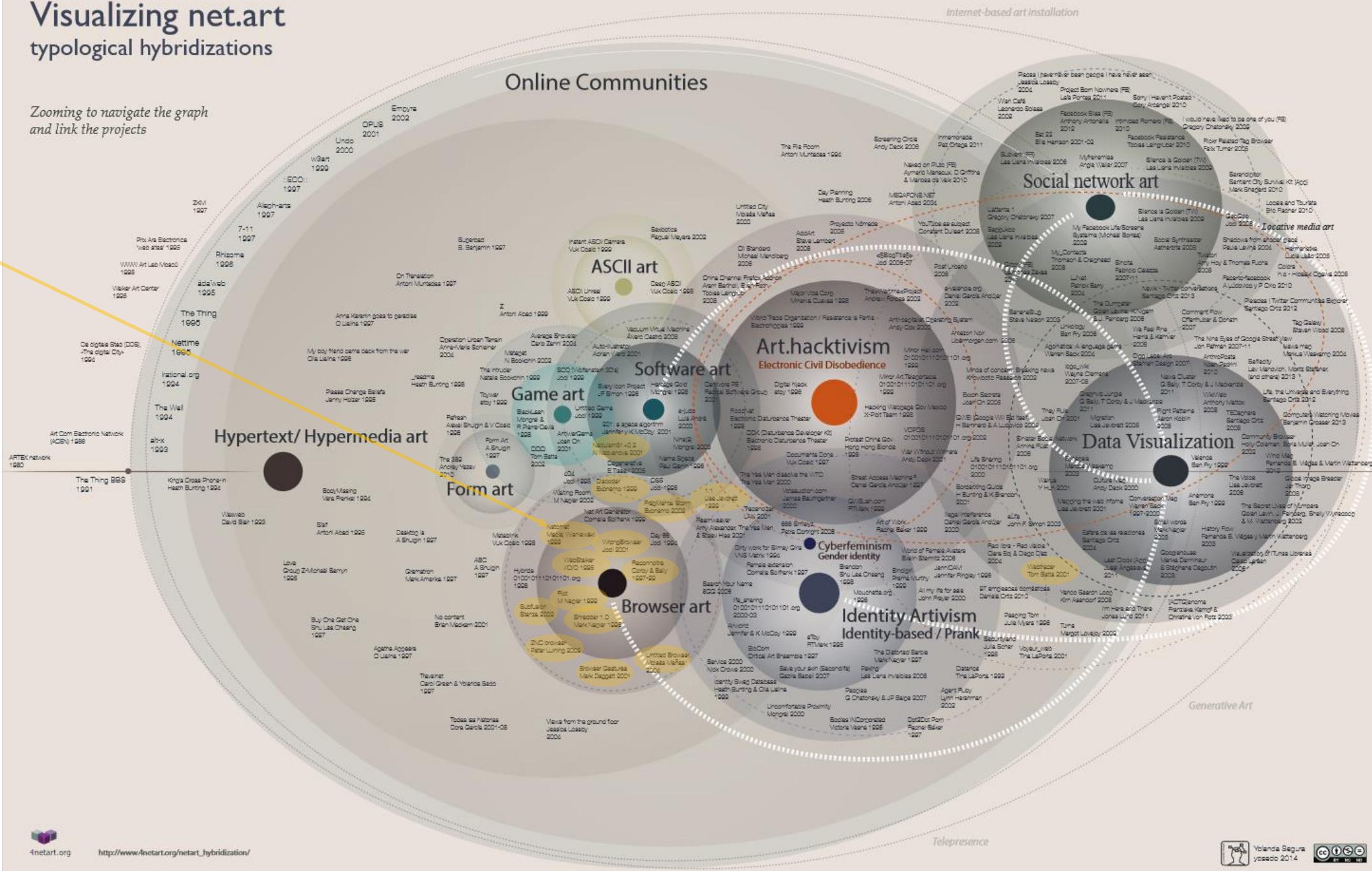
Colin Post: The Art of Digital Curation: Co-operative Stewardship of Net-Based Art. Archivaria, (92), 2001, pp. 6–47.

<https://doi.org/10.7202/1084738ar>

Visualizing net.art typological hybridizations

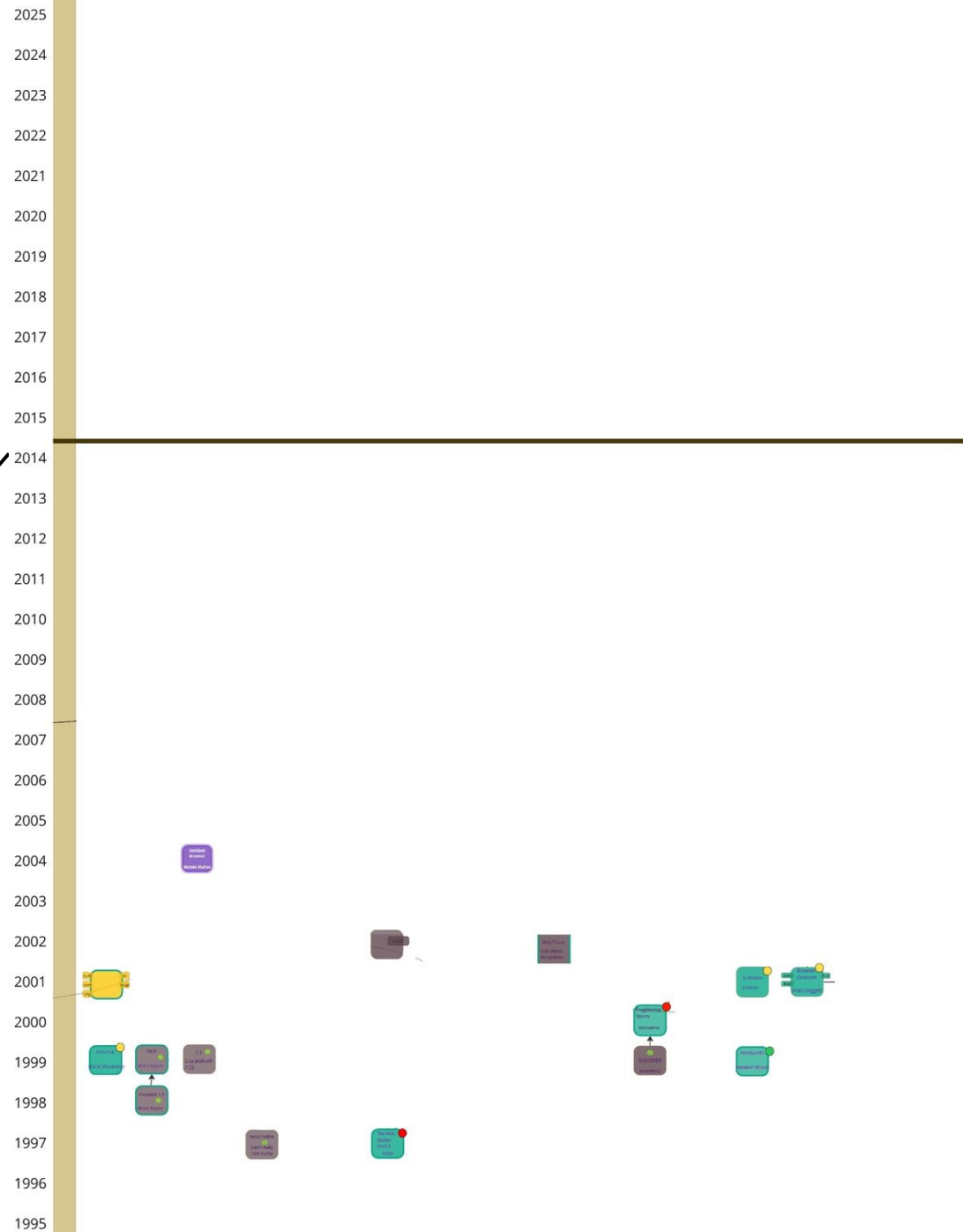
... here is the
state of the
art 2014 in
another
survey
diagram of
net art:
known artistic
browsers: 15

Zooming to navigate the graph
and link the projects

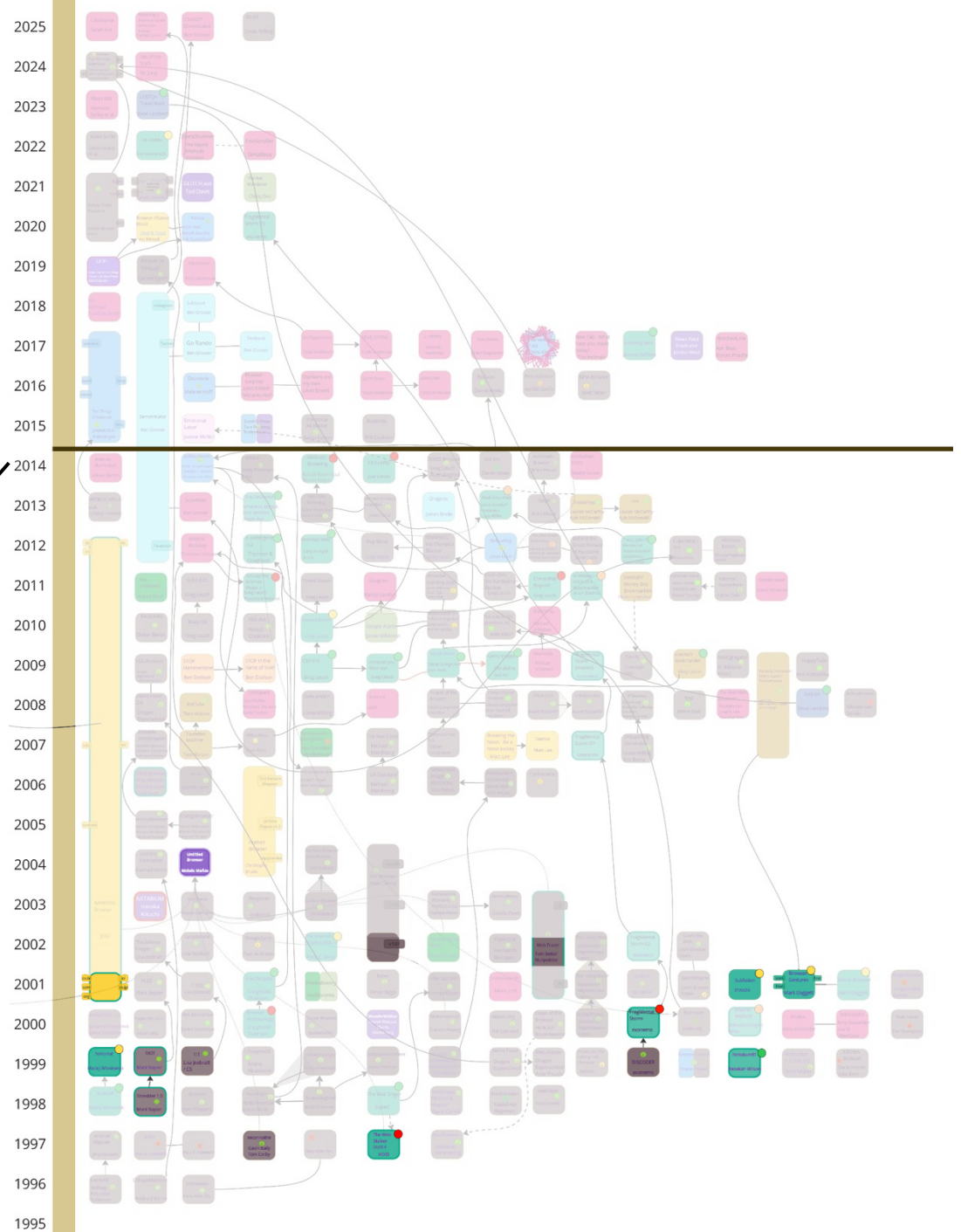


Yolanda
Segura:
Visualizing
net.art, 2014.
Source:
http://www.4netart.com/org/netart_hybridization/hibridaciones_netart_yosedo.pdf

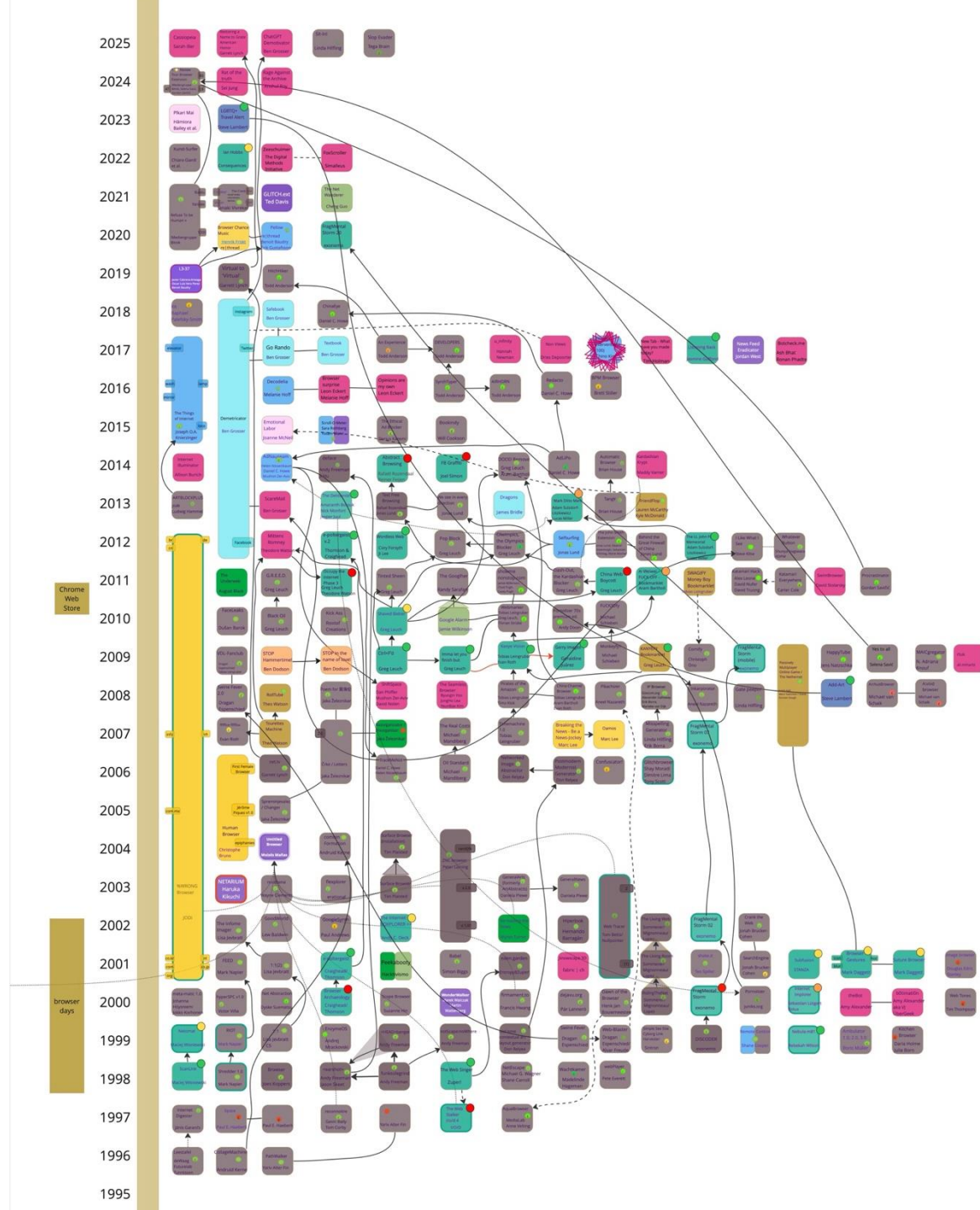
Yolanda Segura:
Visualizing
net.art, 2014.
Source:
http://www.4netart.com/org/netart_hybridization/hibridaciones_netart_yosedo.pdf



Yolanda Segura:
Visualizing
net.art, 2014.
Source:
http://www.4netart.com/org/netart_hybridization/hibridaciones_netart_yosedo.pdf



March 2026



Choose Your Filter!

Browser Art seit den Anfängen
des World Wide Web

Noch bis
24.8.25



Show works as often as possible

Exhibition

Choose Your Filter!

Browser Art
since the
Beginnings of
the World Wide
Web

Sat, February 01
– Sun, August
24, 2025



© 2004 | Center for Art and Media Karlsruhe, photo: Felix Grunschell

software art

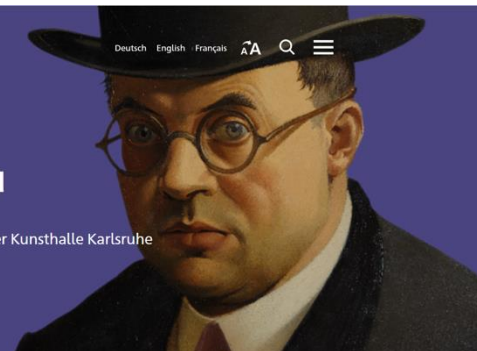
STAATLICHE
KUNSTHALLE
KARLSRUHE

Sammlungspräsentation

ab 29. Apr. 2023

See You

Begegnungen mit der Kunsthalle Karlsruhe



paintings

Show works as little as possible

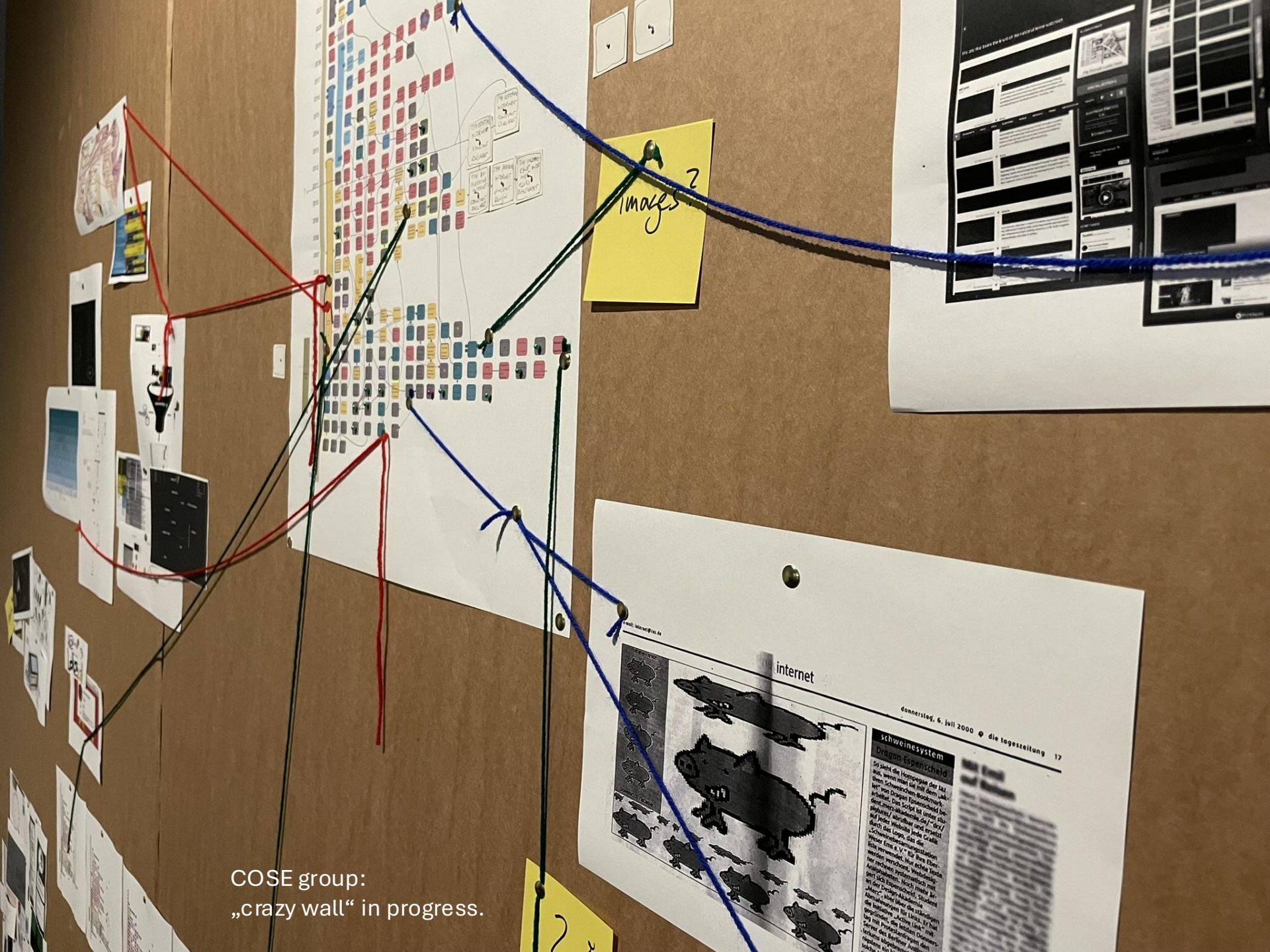
2 simultaneous exhibitions at ZKM 2025 –
2 very different conservation strategies



Choose Your Filter! Browser Art since the Beginnings of the World Wide Web, ZKM, Feb 1-Aug 28, 2025

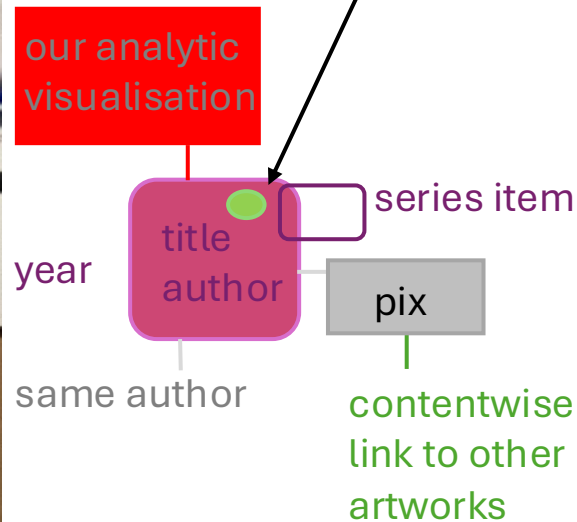
survey of a field

Timeline / Database (audiovisual, poster, **book**)
+ source/reference
(COSE: *Browser Art Browser*, 2025: based on Jeffrey Shaw: *Linear Browser*, 1999)



COSE group:
„crazy wall“ in progress.

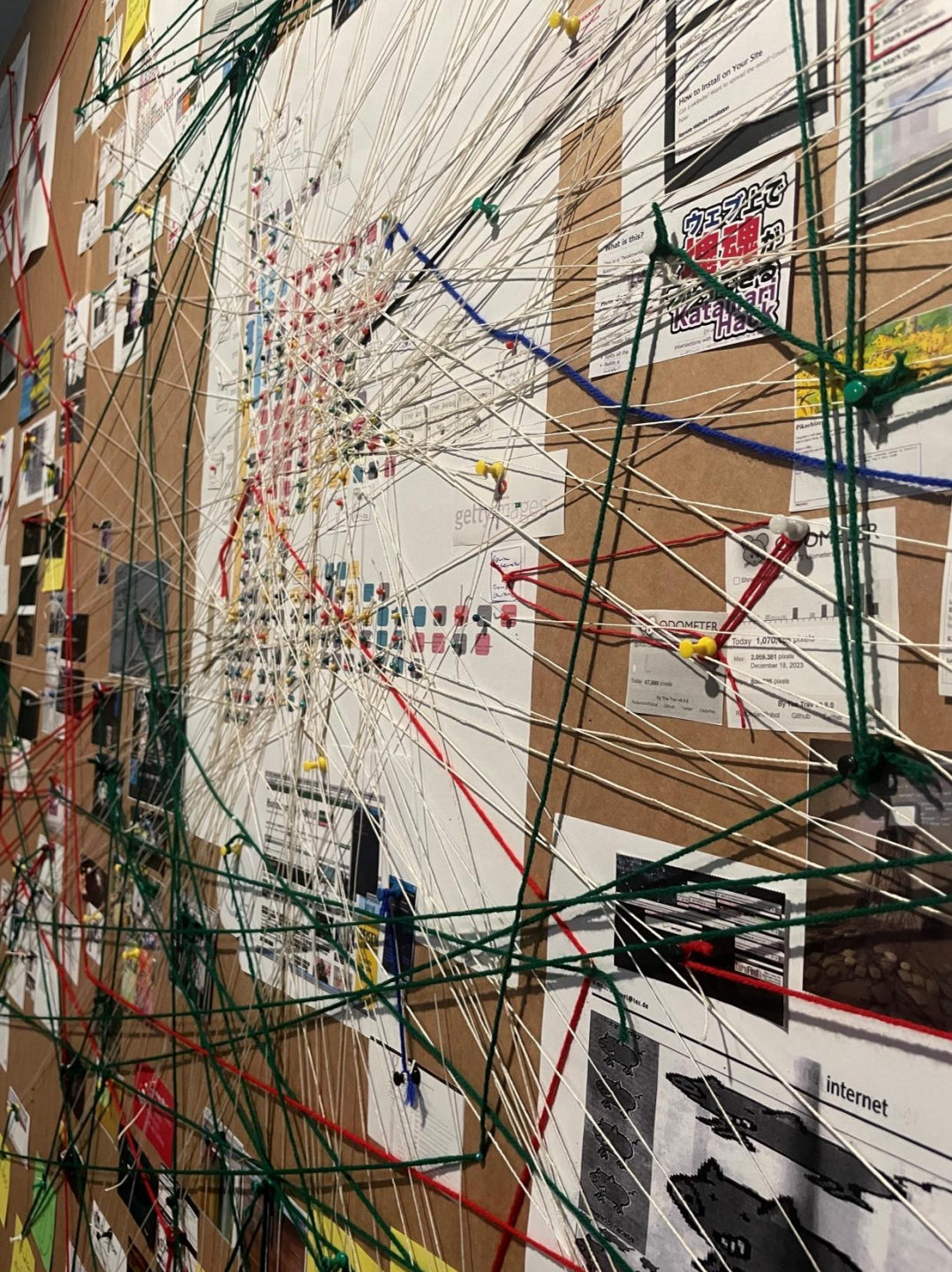
colour code + circle:
status of research (interview)
and of publication
(catalogue entry)

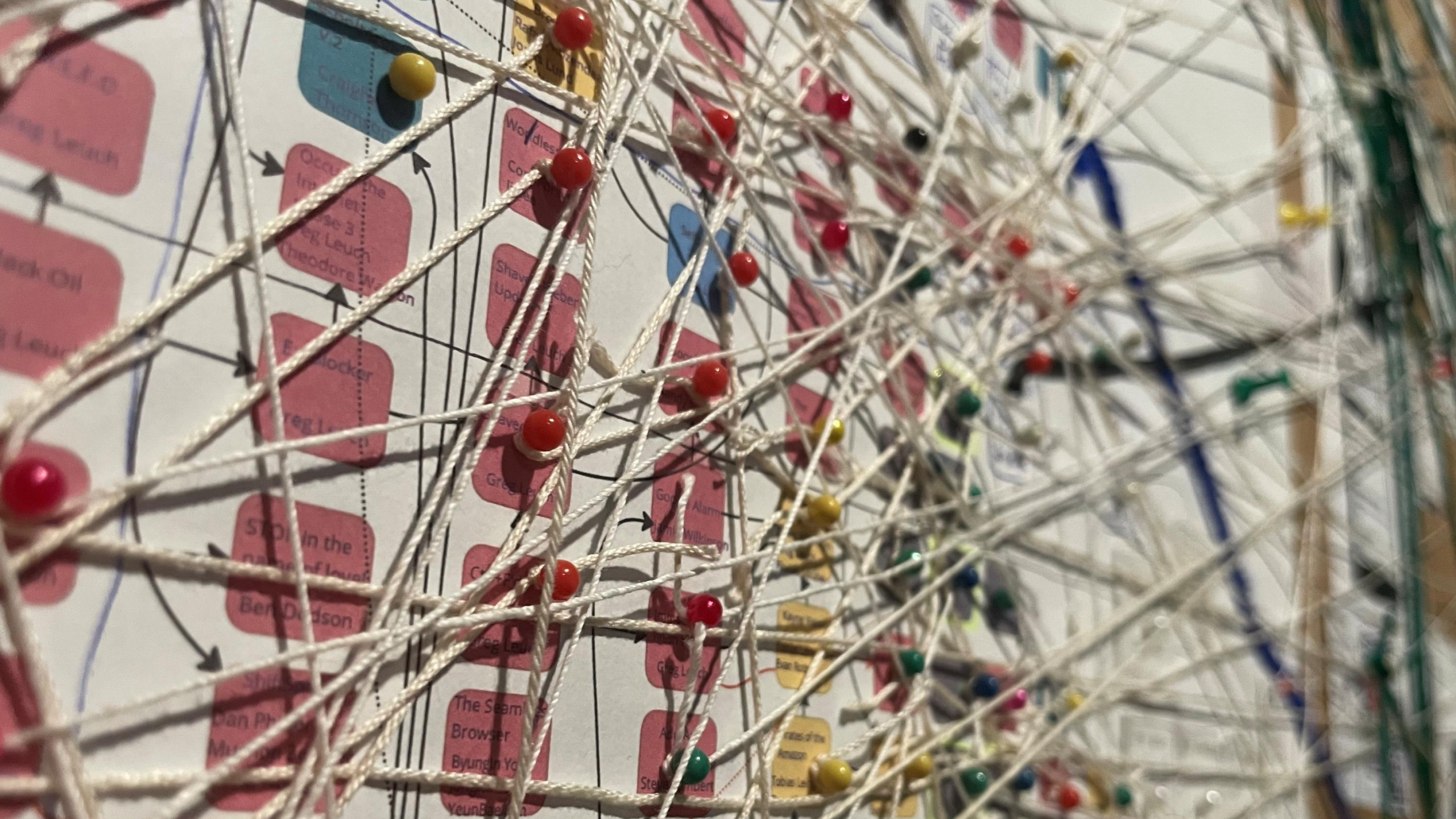


documentation of:
past activities

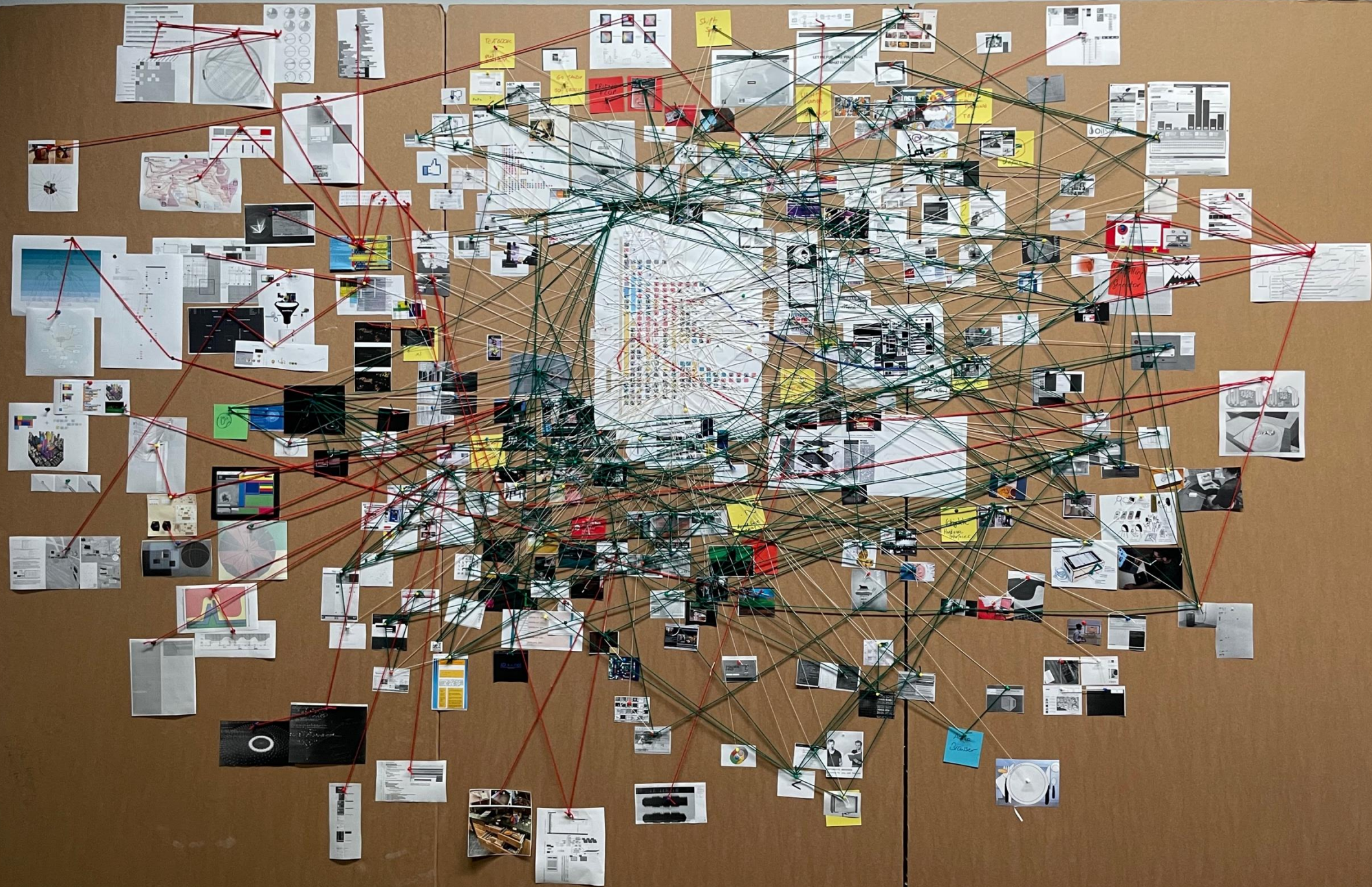
current activities

emergent feats:
future activities









survey of a field

COSE: Crazy wall,
State of the art:
22.1.2026.

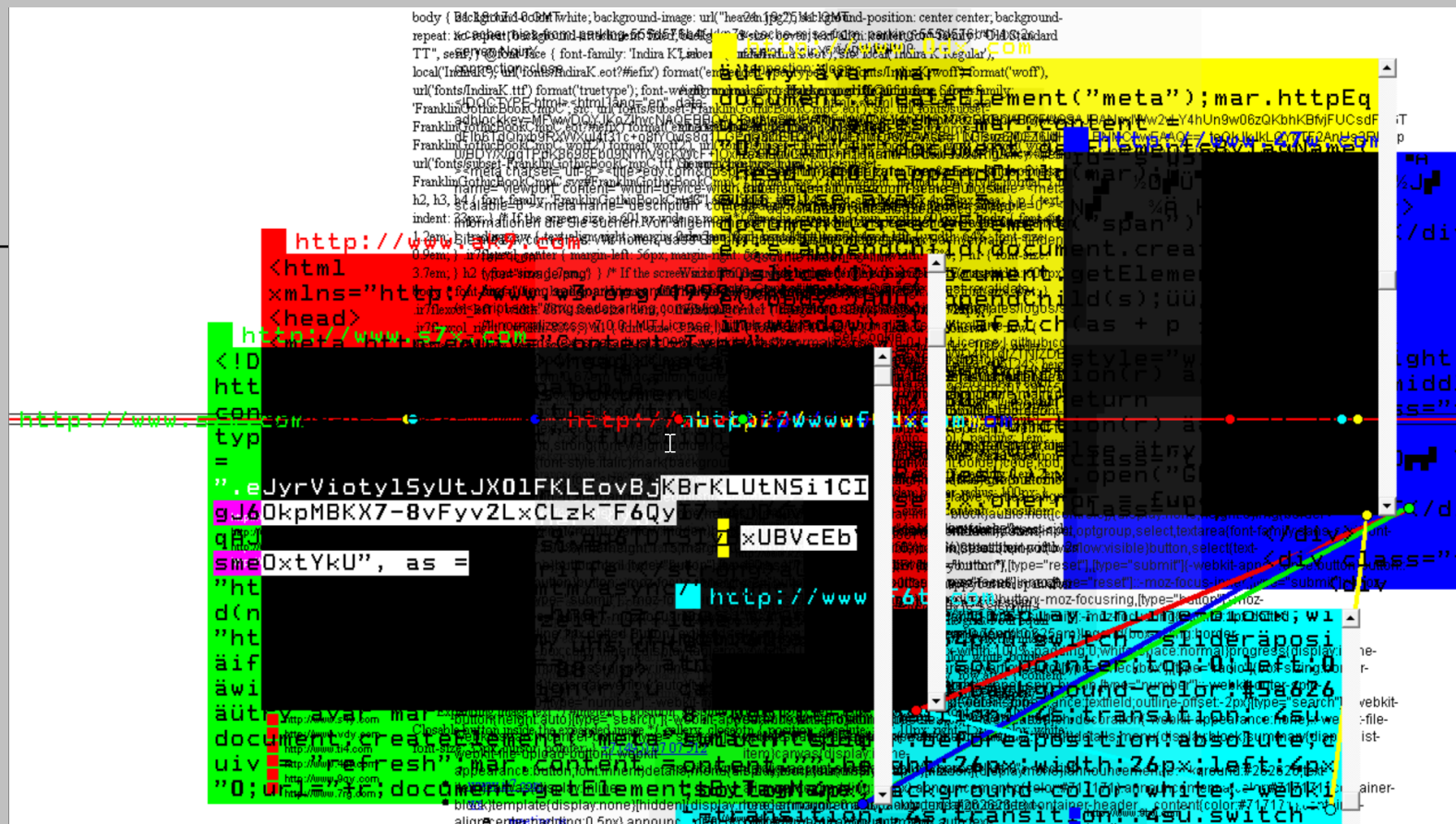
Visualising
semantic fields,
the cross-
referencing and
the catalogue

-
living document

**With about 220
artistic web
browsers we
considerably
expanded what
was known of
that genre ...**

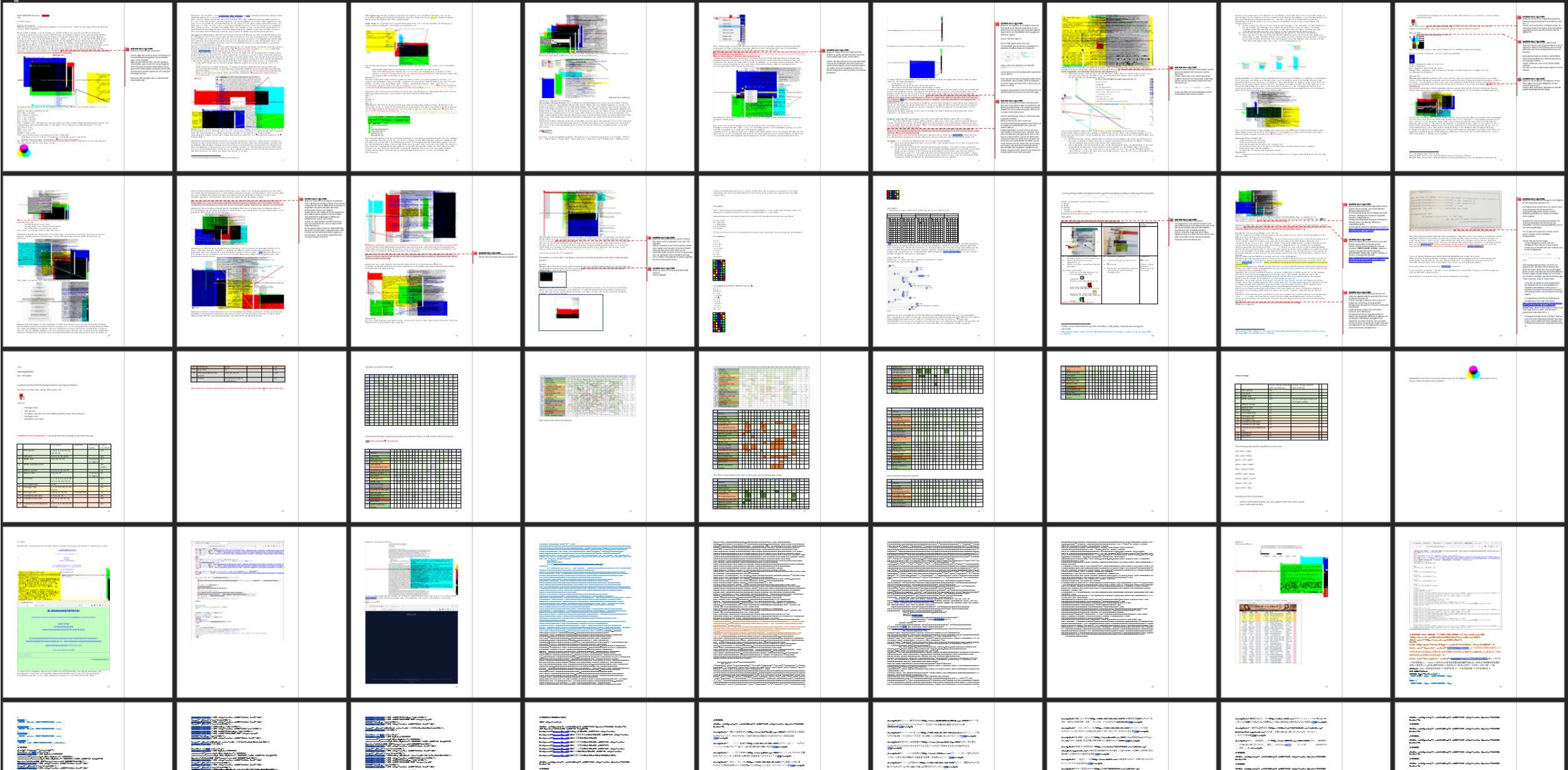
%WRONG Browser

	mac osx	pc win
.br		
.cn		
.co.jp		
.co.kr		
<u>.com</u>		
.com.mx		
.de		
.info		
.nl		
<u>.org</u>		
.us		

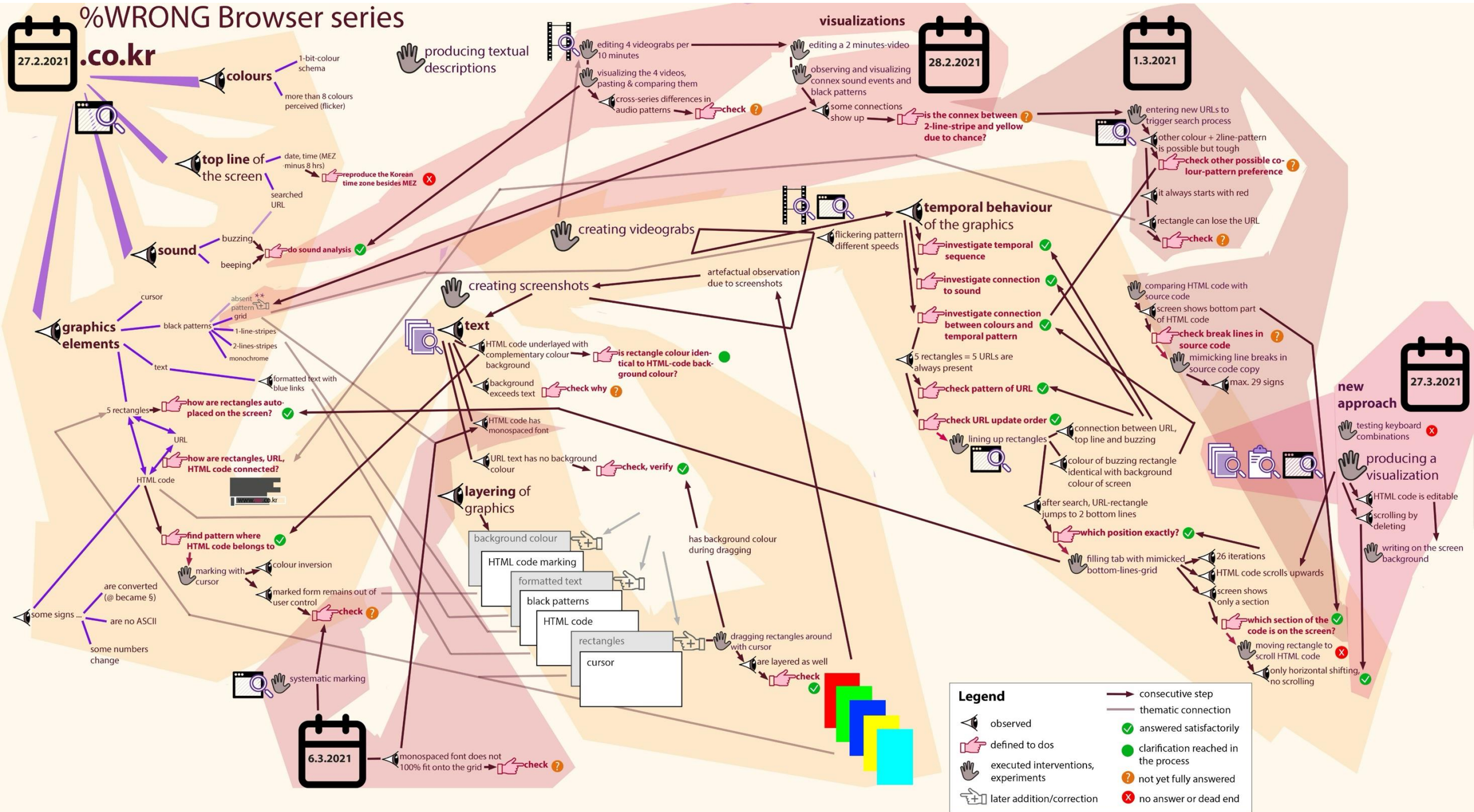


JODI: %WRONG Browser .com. screenshot

JODI: %WRONG Browser series, 2001-2012. Download menu.



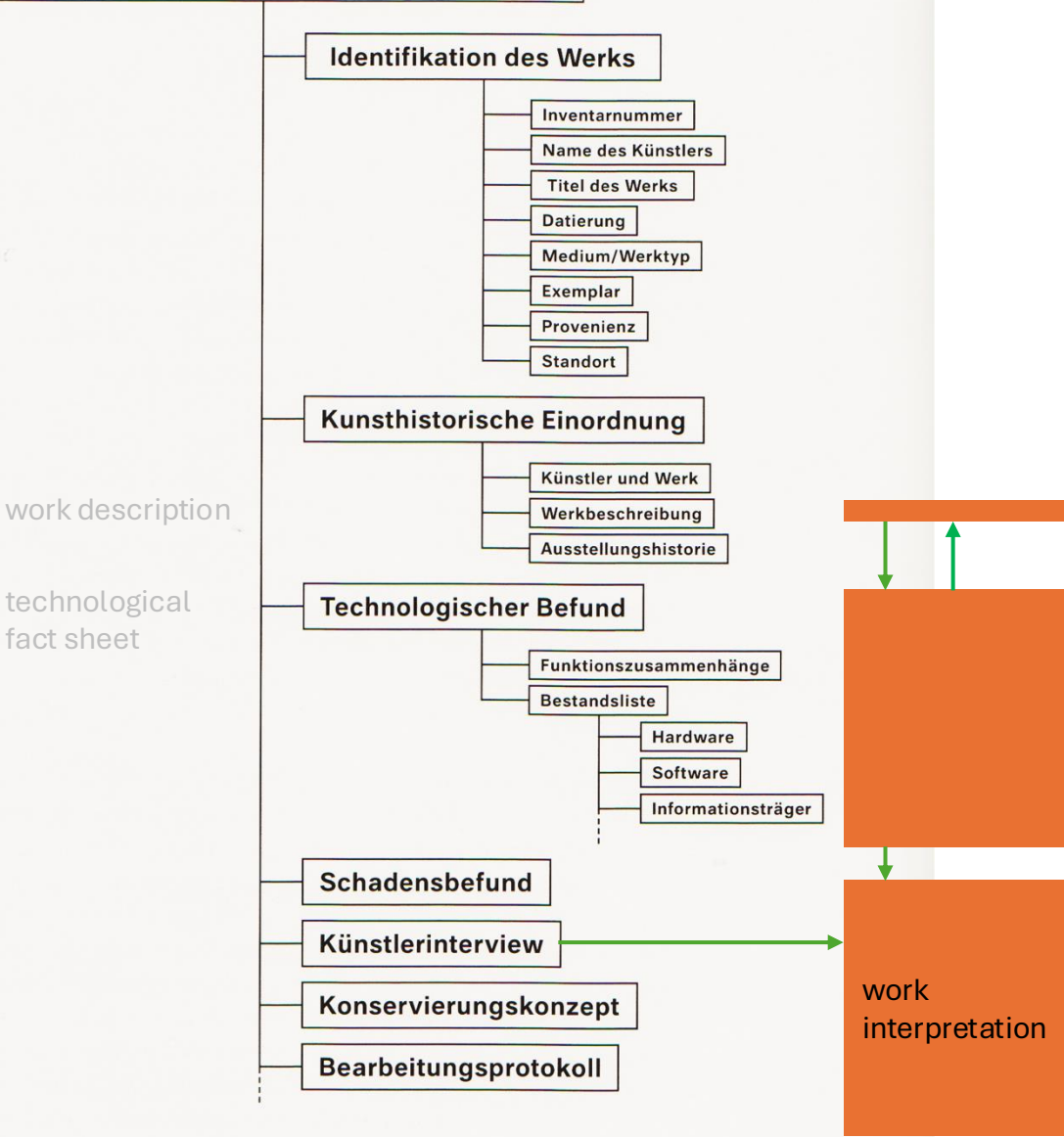
Semi-structured documentation of .com (mix of text, tabs, screenshots, diagrams, comparisons, comments ...) in a word document



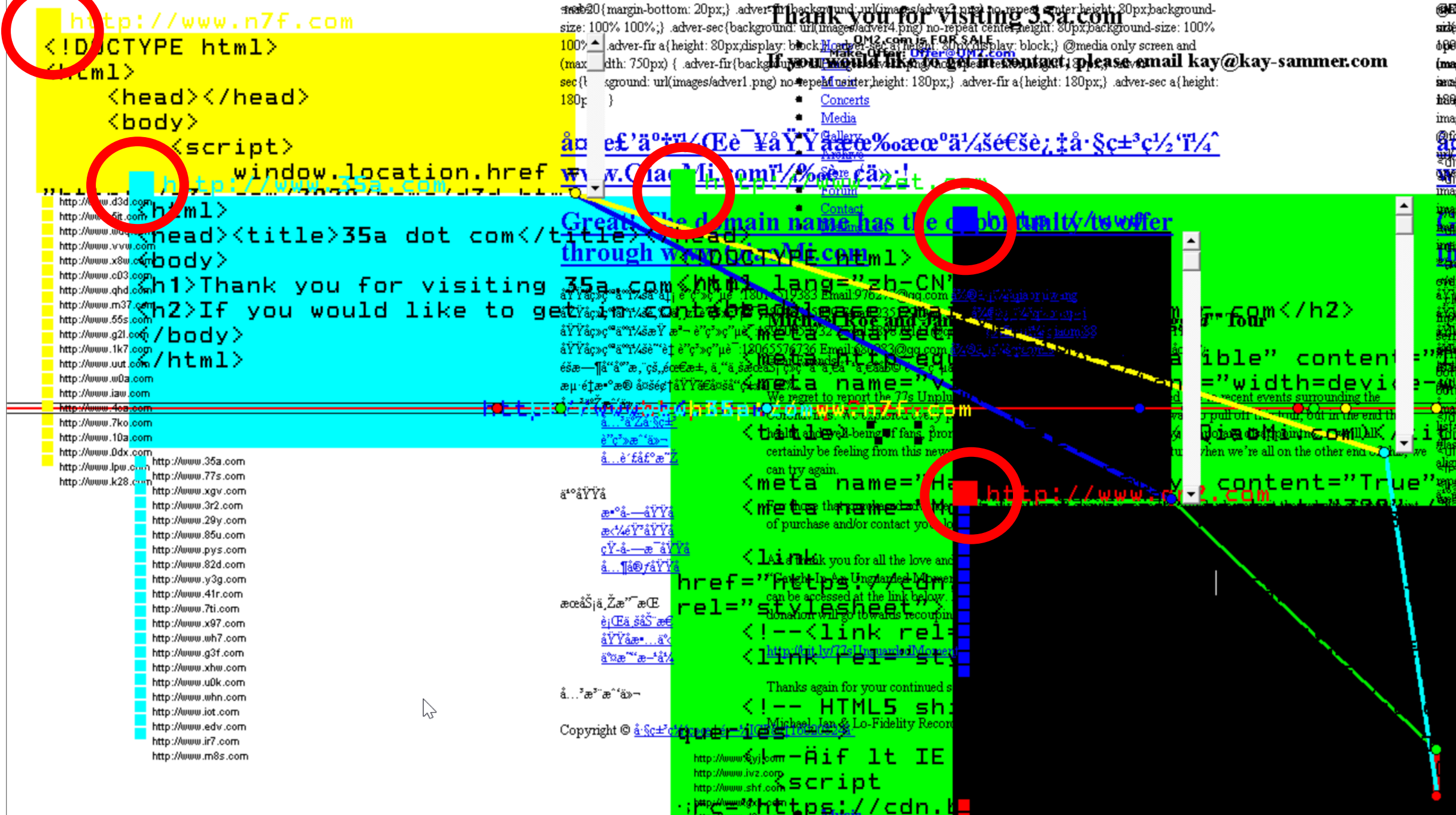


Serexhe, Bernhard (ed.): digital art conservation. Konservierung digitaler Kunst: Theorie und Praxis, Ambra | V / ZKM, Vienna 2013, p. 340.

Konservierungsdokumentation

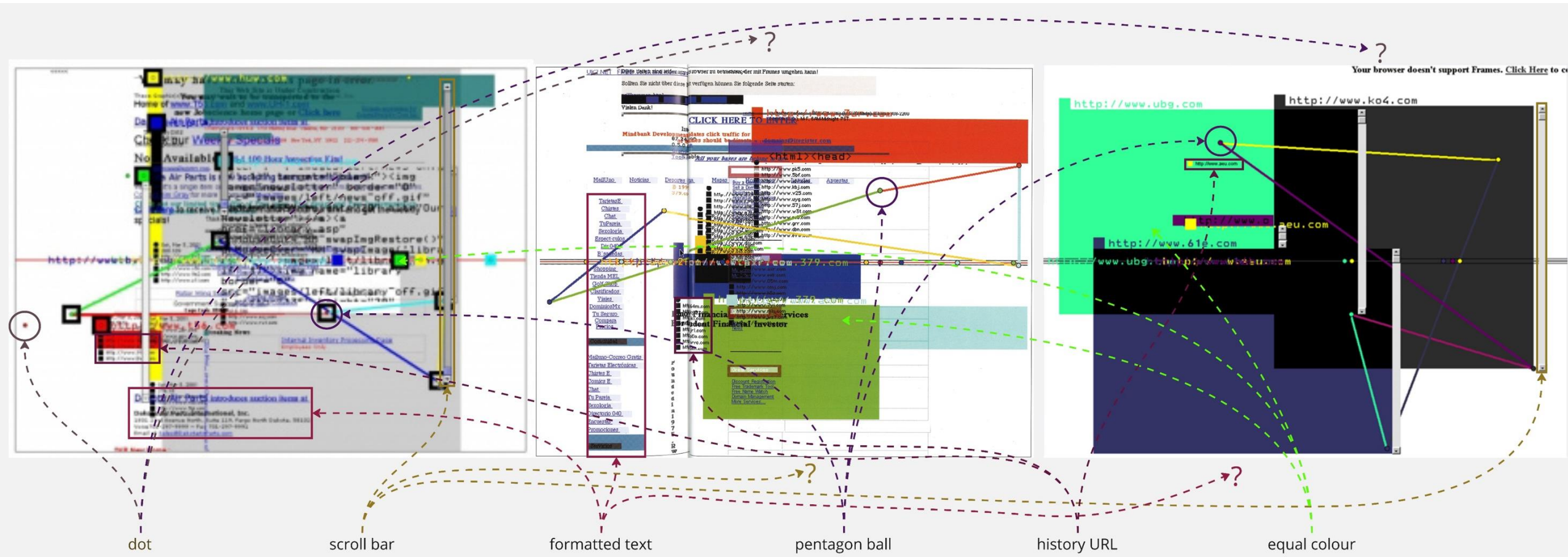


Dokumentationsstruktur zur Aufnahme der für das Projekt digital art conservation ausgewählten Werke
Foto © ZKM | Karlsruhe



JODI: .com, 2001. From the series %WRONG Browser. Still, reworked.

The screenshot dilemma: too quick&easy to be good (enough)

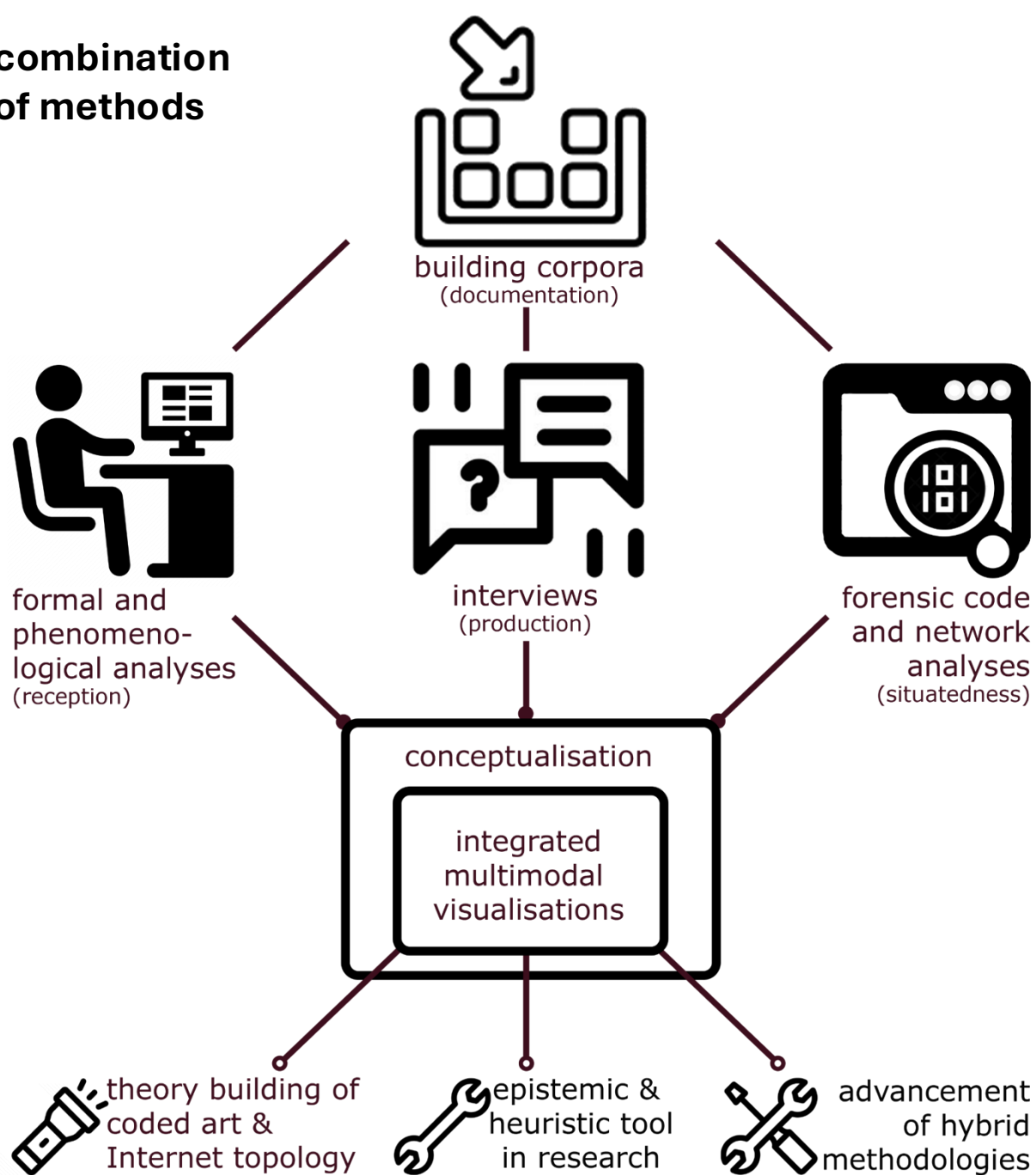


Dekker, Annet: World Wide Wrong, JODI, 24.8.2005, in: <http://aaan.net/world-wide-wrong-jodi/> (28.1.2023).

Baumgärtel, Tilman: net.art 2.0, New Materials Towards Net Art, Verlag für moderne Kunst: Nürnberg 2001, pp. 176-177.

Mackern, Brian: // navegadores web alternativos // v 2.004, in: <https://netart.org.uy/browser-art/> (18.5.2023). The screenshot was taken on November 17, 2001.

combination of methods



Methodology

To provide the first in-depth interpretation of this iconic browser art piece we needed to secure our knowledge base by documenting it thoroughly. We combined:

- **Interviews** (social sciences approaches)
- **Phenomenological analysis** (art history)
- **Forensic code analysis** (digital humanities, computer science, conservation)

We found the use of screenshots as being questionable and developed:

- **Model building** (arts, media design, art history)

Carbon Nanotubes: Synthesis, Integration, and Properties

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Stanford, California 94305

Received January 23, 2002

ABSTRACT

Synthesis of carbon nanotubes by chemical vapor deposition over patterned catalyst arrays leads to nanotubes grown from specific sites on surfaces. The growth directions of the nanotubes can be controlled by van der Waals self-assembly forces and applied electric fields. The patterned growth approach is feasible with discrete catalytic nanoparticles and scalable on large wafers for massive arrays of novel nanowires. Controlled synthesis of nanotubes opens up exciting opportunities in nanoscience and nanotechnology, including electrical, mechanical, and electromechanical properties and devices, chemical functionalization, surface chemistry and photochemistry, molecular sensors, and interfacing with soft biological systems.

Introduction

Carbon nanotubes represent one of the best examples of novel nanostructures derived by bottom-up chemical synthesis approaches. Nanotubes have the simplest chemical composition and atomic bonding configuration but exhibit perhaps the most extreme diversity and richness among nanomaterials in structures and structure–property relations.¹ A single-walled nanotube (SWNT) is formed by rolling a sheet of graphene into a cylinder along an (m,n) lattice vector in the graphene plane (Figure 1). The (m,n) indices determine the diameter and chirality, which are key parameters of a nanotube. Depending on the chirality (the chiral angle between hexagons and the tube axis), SWNTs can be either metals or semiconductors, with band gaps that are relatively large (~ 0.5 eV for typical diameter of 1.5 nm) or small (~ 10 meV), even if they have nearly identical diameters.¹ For same-chirality semiconducting nanotubes, the band gap is inversely proportional to the diameter. Thus, there are infinite possibilities in the type of carbon tube “molecules”, and each nanotube could exhibit distinct physical properties.

The past decade has witnessed intensive theoretical and experimental effort toward elucidating the electronic sensitivity of the electronic properties of nanotubes to their atomic structures.^{2–4} It has been revealed that metallic and semiconducting nanotubes can be synthesized by arc-discharge, laser ablation, and chemical vapor deposition methods. Metallic SWNTs have become model systems for investigating the rich quantum phenomena in quasi-1d solids, including single-electron

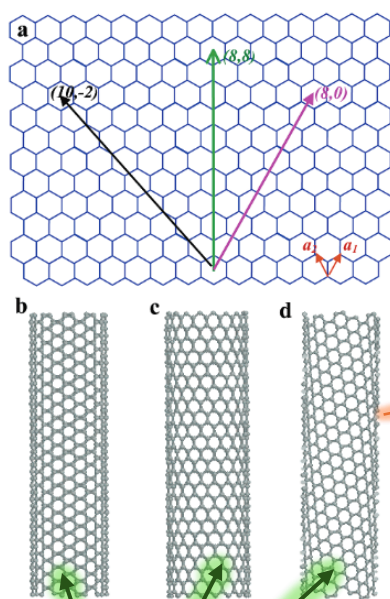


FIGURE 1. (a) Schematic honeycomb structure of a graphene sheet. Single-walled carbon nanotubes can be formed by folding the sheet along lattice vectors. The two basis vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ are shown. Folding of the (8,8), (8,0), and (10,-2) vectors lead to armchair (b), zigzag (c), and chiral (d) tubes, respectively.

charging, Luttinger liquid, weak localization, ballistic transport, and quantum interference.^{2–4,7,8} On the other hand, semiconducting nanotubes have been exploited as novel building blocks for nanoelectronics, including transistors and logic, memory, and sensory devices.^{2–4}

Nanotube characterization and device explorations have been greatly facilitated by progress in nanotube synthesis and characterization. In the past few years, many aspects of basic research and practical application requirements have been defined, and motivating synthetic methods for nanotubes and homogeneous materials. This complementary cycle continues, and it is now at a time when nanotube synthesis has evolved from enabling growth into consciously controlling the growth. While the extreme sensitivity of nanotube structure–property relations has led to rich science and promises a wide range of applications, it poses a significant challenge to chemical synthesis in controlling the nanotube diameter and chirality. Understanding how to synthesize nanotubes with predictable

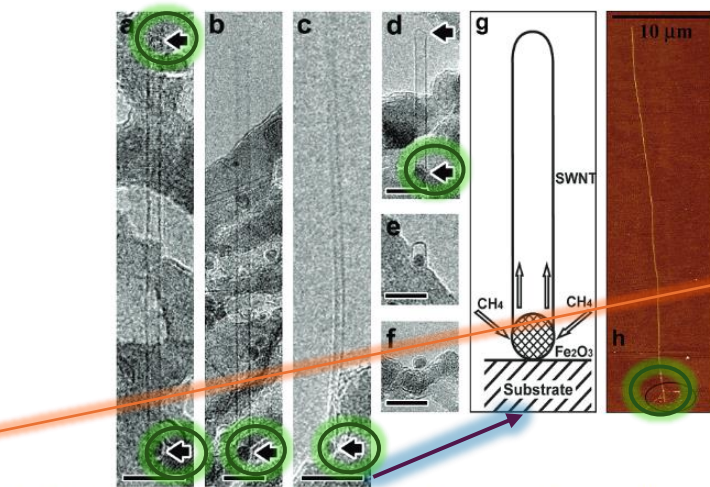


FIGURE 3. (a–f) TEM images of SWNTs grown from discrete nanoparticles, showing particle–nanotube relationships. Scale bars: 10 nm. (a–d) SWNTs grown from discrete nanoparticles (dark dots at the bottom of the images). The arrows point to the ends of the nanotubes. The ends extending out of the membrane in (b) and (c) are not imaged due to thermal vibration. The background roughness reflects the TEM grid morphology. (e) Image of an ultrashort (~ 4 nm) nanotube capsule grown from a ~ 2 nm nanoparticle. (f) Image of a nanoparticle surrounded by a single graphitic shell. (g) A schematic model for nanotube growth. (h) AFM image of a 50 μm long SWNT grown from nanoparticles patterned into the circled region.

explored patterned growth of SWNTs on full 4-in. SiO₂/Si wafers.²⁴ We have also demonstrated that catalyst islands with high density can be grown by a photolithography approach (Figure 5a). We have also investigated both CVD and H₂ and H₂ co-flow) affect the growth of SWNTs. This aspect has been systematically studied in our earlier synthesis work. We have identified that, for relatively low CH₄ flow rates, the CVD process is ultrasensitive to the amount of H₂ co-flow, undergoing pyrolysis, growth, and inactive reaction regimes with increased H₂ addition. This understanding has enabled us to grow high-quality SWNTs (Figure 4b) from massive arrays of well-defined surface sites on full 4-in. wafers.²⁴

Electrical Properties and Interplay with Mechanical and Chemical Properties

Patterned growth of carbon nanotubes on substrates has allowed for convenient, simple, and controlled integration of nanotubes into various device structures for elucidating the properties of individual molecular wires. With these devices, we have carried out electron transport measurements of semiconducting, quasi-metallic, and metallic SWNTs grown by CVD. Suspended nanotubes have also been integrated into addressable structures for mechanical

(a) **Electrical Properties of Nanotubes.** We have revealed that the growth of SWNTs on catalyst islands with high density can be grown by a photolithography approach (Figure 5a). We have also investigated both CVD and H₂ and H₂ co-flow) affect the growth of SWNTs. This aspect has been systematically studied in our earlier synthesis work. We have identified that, for relatively low CH₄ flow rates, the CVD process is ultrasensitive to the amount of H₂ co-flow, undergoing pyrolysis, growth, and inactive reaction regimes with increased H₂ addition. This understanding has enabled us to grow high-quality SWNTs (Figure 4b) from massive arrays of well-defined surface sites on full 4-in. wafers.²⁴

The second type of observed SWNTs synthesized by CVD appears to be quasi-metallic with small band gaps on the order of 10 meV.²⁴ These nanotubes are not as sensitive as semiconducting SWNTs to electrostatic doping by gate voltages but exhibit a conductance dip associated with the small band gap. These nanotubes correspond to

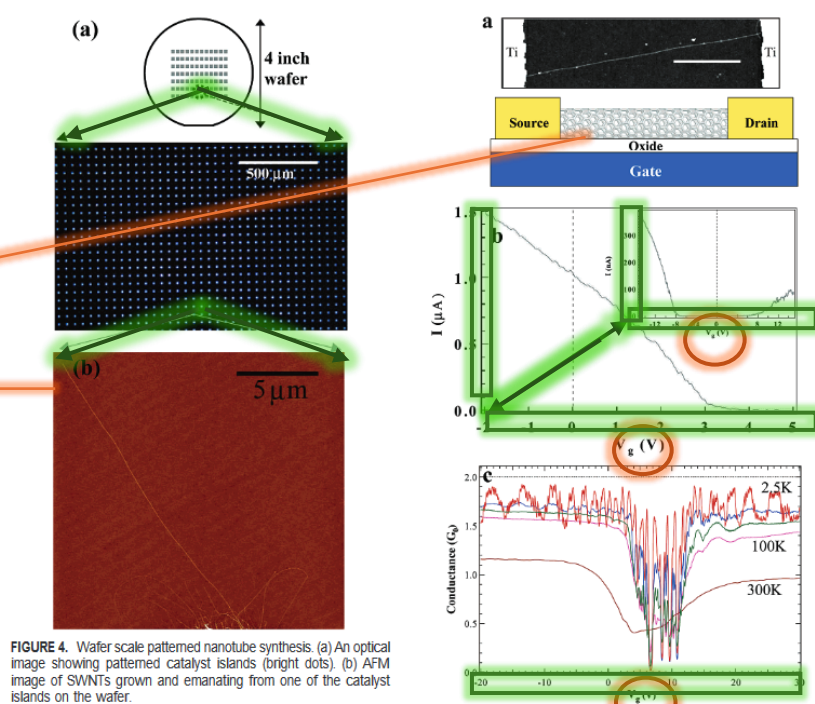


FIGURE 4. Wafer scale patterned nanotube synthesis. (a) An optical image showing patterned catalyst islands (bright dots). (b) AFM image of SWNTs grown and emanating from one of the catalyst islands on the wafer.

gap is slightly smaller than the band gap of the nonflat nature of the nanotube surface.²⁵ Temperature-dependent measurements for some of the quasi-metallic-like SWNTs (room-temperature resistance ~ 10 – 20 k Ω) reveals increased electrical conductance at low temperatures, reaching the $4e^2/h = 2G_0$ quantum conductance (6.45 k Ω in resistance) limit at ~ 1.5 K (Figure 5c). Quantum interference effects have also been observed.⁸ The results suggest that (1) phonon is one of the fundamental scattering mechanisms in SWNTs at room temperature, (2) excellent ohmic contacts can be made to nanotubes with transmission probability $T \approx 1$, and (3) electron transport is highly phase coherent and ballistic in nanotubes at low temperatures.⁸

Truly metallic armchair SWNTs are rare in our CVD-grown nanotubes. The conductance of this type of nanotube is the least sensitive to gate voltages. With highly transparent ohmic contacts to these nanotubes, we have observed near $4e^2/h$ quantum conductance in SWNTs as long as 4 μm at low temperatures (J. Kong and H. Dai, unpublished results), suggesting a long mean free path

FIGURE 5. Electrical properties of individual nanotubes. (a) AFM image of a SWNT contacted by two Ti electrodes and a schematic cross-section view of the device. Scale bar: 1 μm . A gate voltage (V_g) can be applied to electrostatically couple to the nanotube and shift the Fermi level. (b) I – V_g curve for a typical as-made semiconducting SWNT showing p-type FET characteristics. Inset: The p-type FET evolves into a nearly intrinsic semiconductor after removal of surface-adsorbed oxygen. (c) Conductance G (in units of quantum conductance, $G_0 = 2e^2/h$) vs V_g (Fermi energy) for a quasi-metallic SWNT at various temperatures. The conductance approaches the theoretical limit $2G_0$ at low temperatures, with conductance fluctuations due to quantum resonance effects.

for ballistic electron transport in high-quality/perfection CVD-grown SWNTs.

(b) **Nanotube Properties and Devices.** Patterned growth has been used to synthesize suspended SWNTs across trenches with the nanotubes wired up electrically with relative ease.²⁷ We have manipulated a suspended nanotube by using an AFM tip while monitoring its electrical conductance (Figure 6a,b). By so doing, we are able to address how mechanical deformation affects the electrical properties of carbon

Dai, Hongjie: Carbon Nanotubes: Synthesis, Integration, and Properties, in: Acc. Chem. Res., vol. 35, 2002, pp. 1035–1044.

Credo: rethinking of science communication is needed. Art history needs to create visualisations/models – to be more precise, to communicate better (like STEM: how to tell micro-stories) and in multiple ways. These models can evolve into a variety of forms, depending on q&a, and this broadens the accessibility for different interests and publics.

Hongjie Dai was born in Shaoyang, Hunan, P. R. China, in 1966. He received his B.S. from Tsinghua University in 1989, his M.S. from Columbia University in 1991, and his Ph.D. from Harvard University with Charles Lieber in 1994. After postdoctoral work with Richard E. Smalley at Rice University, he joined the faculty of the Department of Chemistry, Stanford University, in 1995. He is currently an Associate Professor of Chemistry and a member of the Stanford Nano Initiative. His research interests are in the synthesis, integration, and properties of carbon nanotubes and other low-dimensional materials.

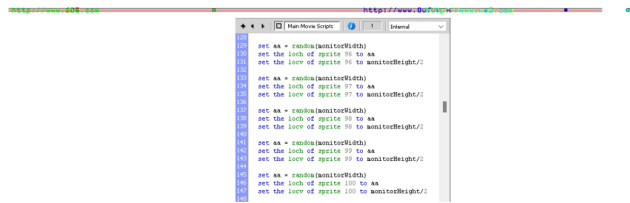


Fig. 6: A reworked screenshot of the initial middle-bar-URL-balls' position (6a). Below the screenshot is the corresponding code from the Main Movie Scripts (6b).

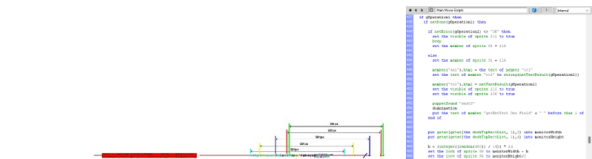


Fig. 7: A reworked screenshot demonstrating the distance between the middle-bar-URL-balls and the middle-bar-URL-text, it can be observed by comparing the colour overlap of the two types of elements (7a). Corresponding code from the Main Movie Scripts (7b).

4) Position of the middle-bar-URL-ball following a new URL-call

Similar to the URL-square [12], the reshuffled positions of the middle-bar-URL-balls [13] follow a different random-function calculation. When a new URL-call is initiated, the horizontal position of a middle-bar-URL-ball is calculated as:

$(\text{monitor width}) - (a \text{ random integer between } 0 \text{ and } 400 \text{ that is also a multiple of } 14)$

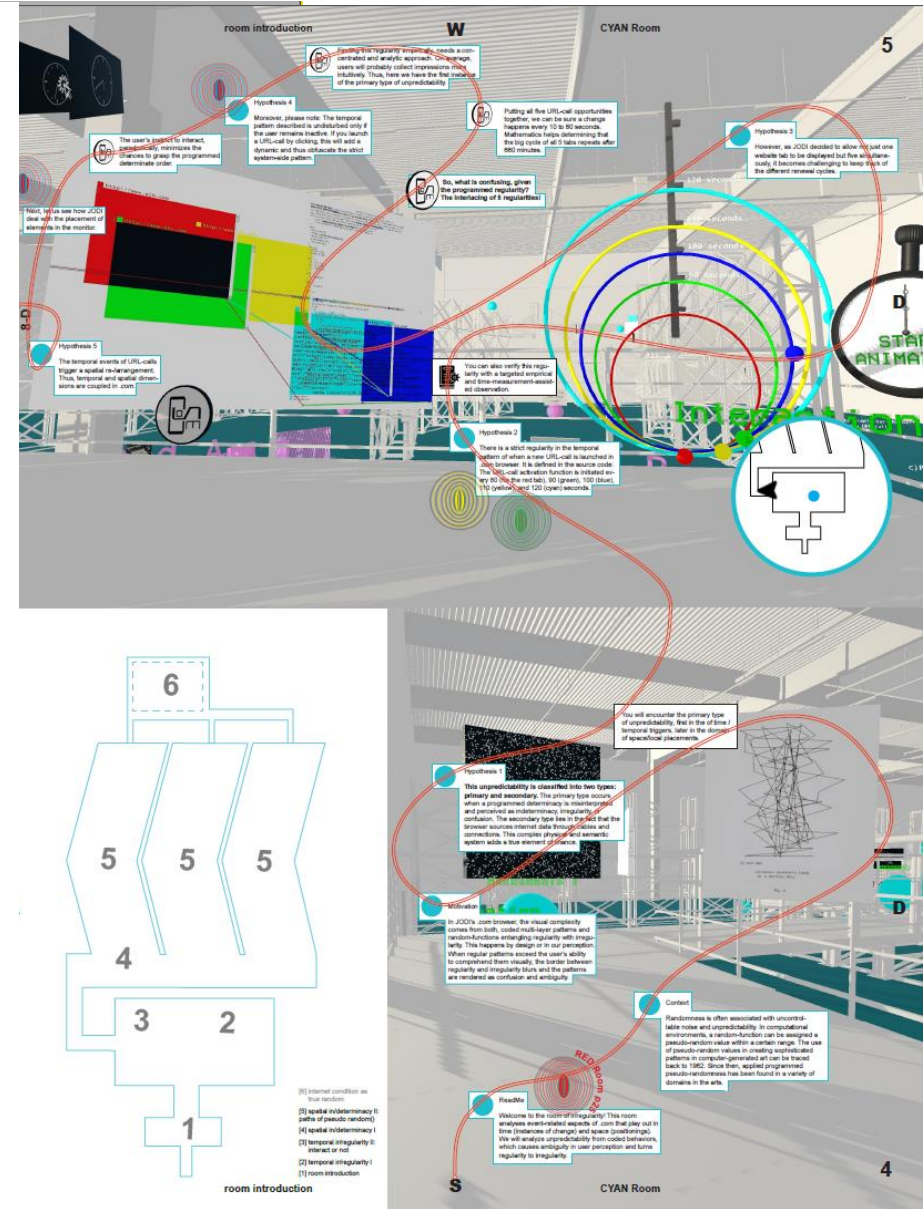
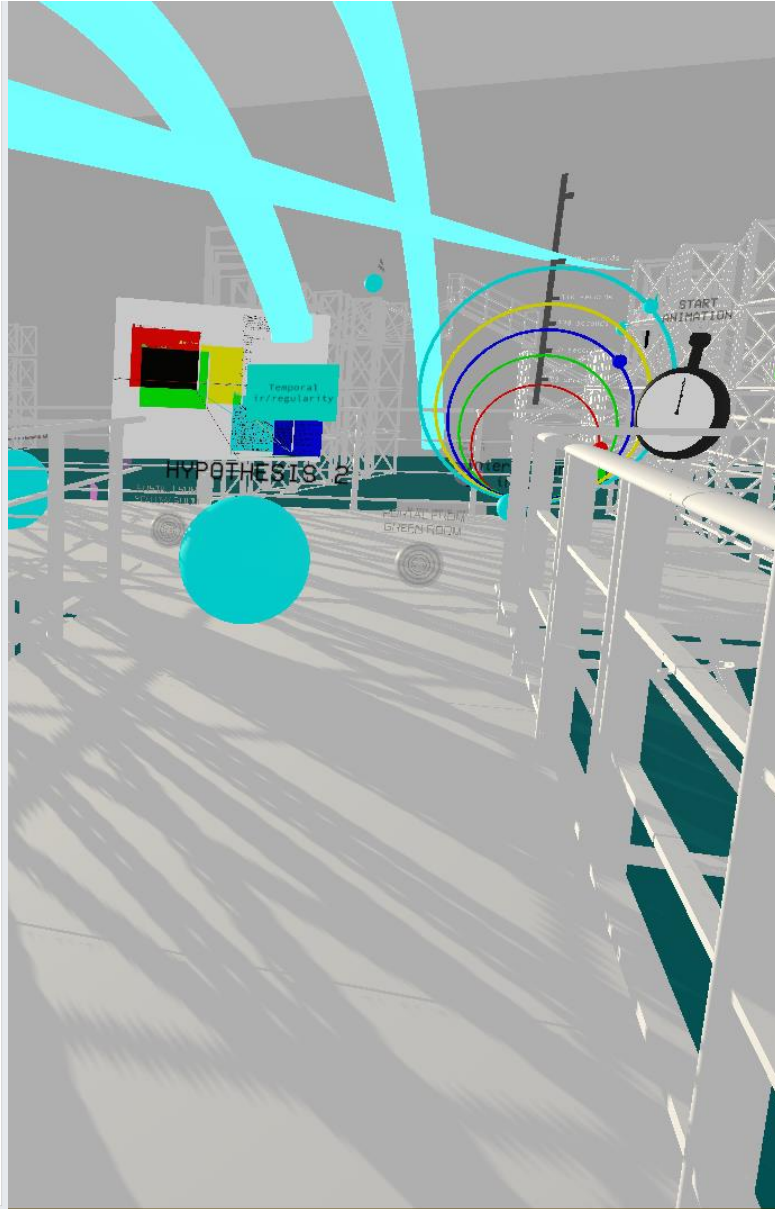
With a standard full HD screen, k being the value of the horizontal position, and 14j the random integer between 0 and 400, then $k=1920-14j$, $1520 < k < 1920$. In this case, the middle-bar-URL-ball for the new URL-call will appear only between the 1520 pixels to 1920 pixels area of the screen as marked in magenta for Fig. 8a.



Fig. 8: A reworked screenshot after all the middle-bar-URL-balls have been reshuffled from their initial positions, the magenta rectangle marking the possible area for middle-bar-URL-ball to settle set by the random-function in a full HD size screen (8a). Corresponding code from the Main Movie Scripts (8b).

The moment the middle-bar-URL-ball's [13] horizontal position is set, the position of the formatted-text ([3], "t1") is determined as well. While the formatted-text always starts at the top of the monitor, its left border is 576 pixels to the left of the corresponding coloured middle-bar-URL-ball. Due to the middle-bar-URL-balls' restricted settling area (Fig. 8a), the formatted-texts are also displayed on the right side of the screen. This provides an

43



In Depth: Roaming Around the Conceptual Space of JODI's .com Browser book

ROAMING.COM game

ROAMING.COM printed game guide

... and the resulting output: COSE's triple publication (2025/2026) + exhibition



ROAMING.COM, 2025.
Unity Game,
Analytic deep-dive

Anything specific about it?

Yes. The programming environment Director retains many features a filmic or theatric logic would imply. Labels like the 'Main Movie Scripts' evoke an orchestration of different bits and pieces. The latter are strung as if they were on a timeline of a video editing tool, except that the linearity can be broken up by using commands with revisions on that 'score.'

ROAMING.COM COSE

Director's Slang. The programming environment, its language and their key command

JODI wrote their browsers in the multimedia authoring software Macromedia Director with the programming language Lingo. Lingo is an object-oriented scripting language native to this programming environment.

Read Me

Net.art.works come to life when 'executed'. Combining static and dynamic code analysis

Welcome to the code forensics! In this room of dissection, we are going to use several forensic tools to find out more information about .com than what can be seen on our monitor while running the browser.

exe

0:00 / 3:25



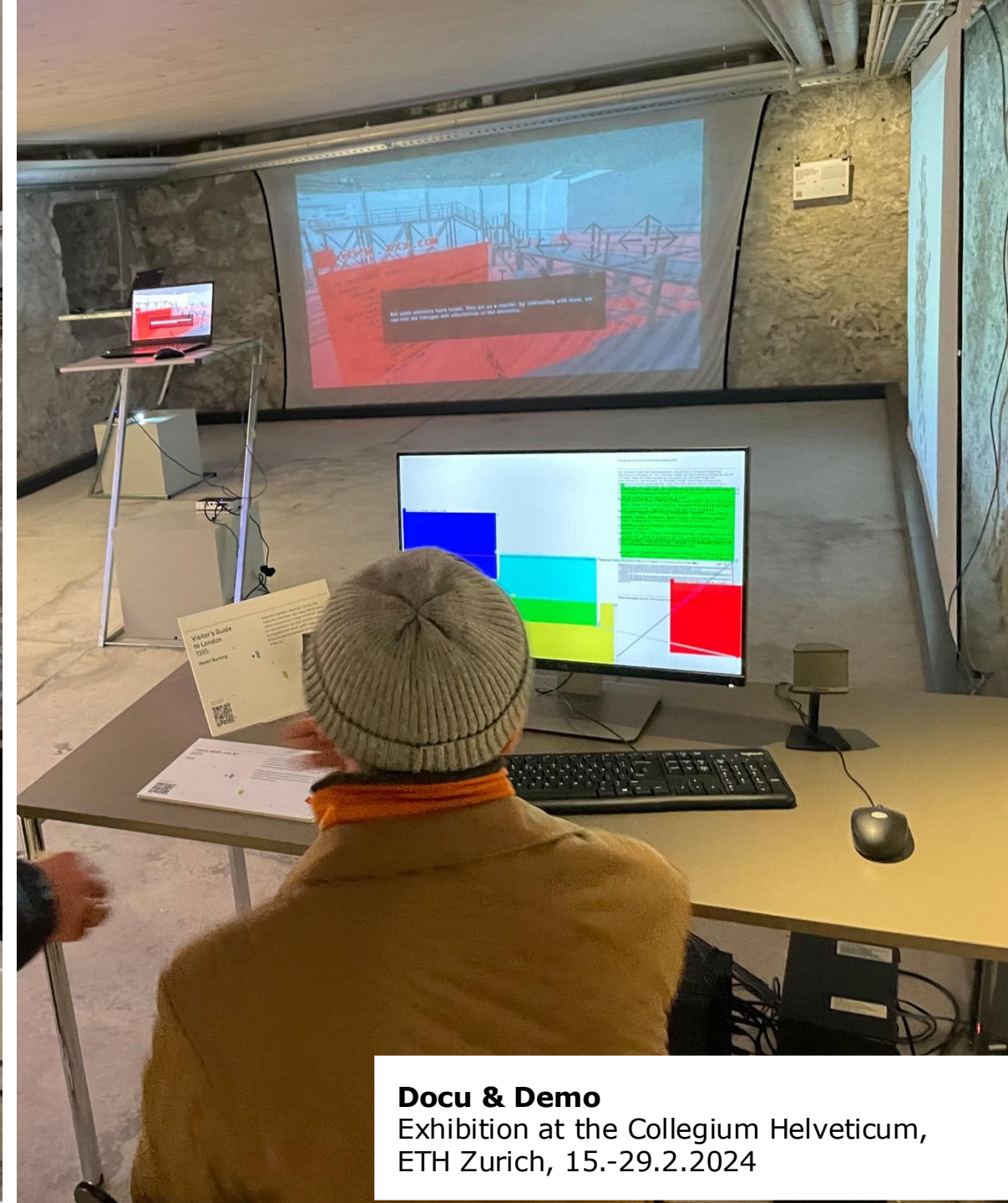
Third Mission



COSE: ROAMING.COM game guide version, 2025.

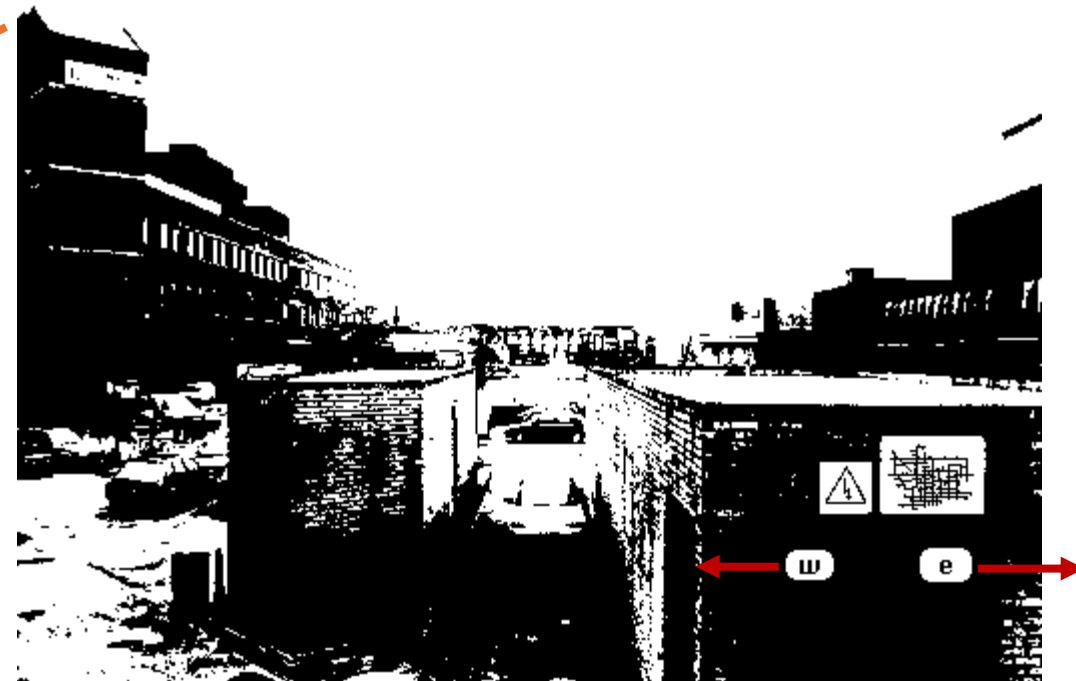
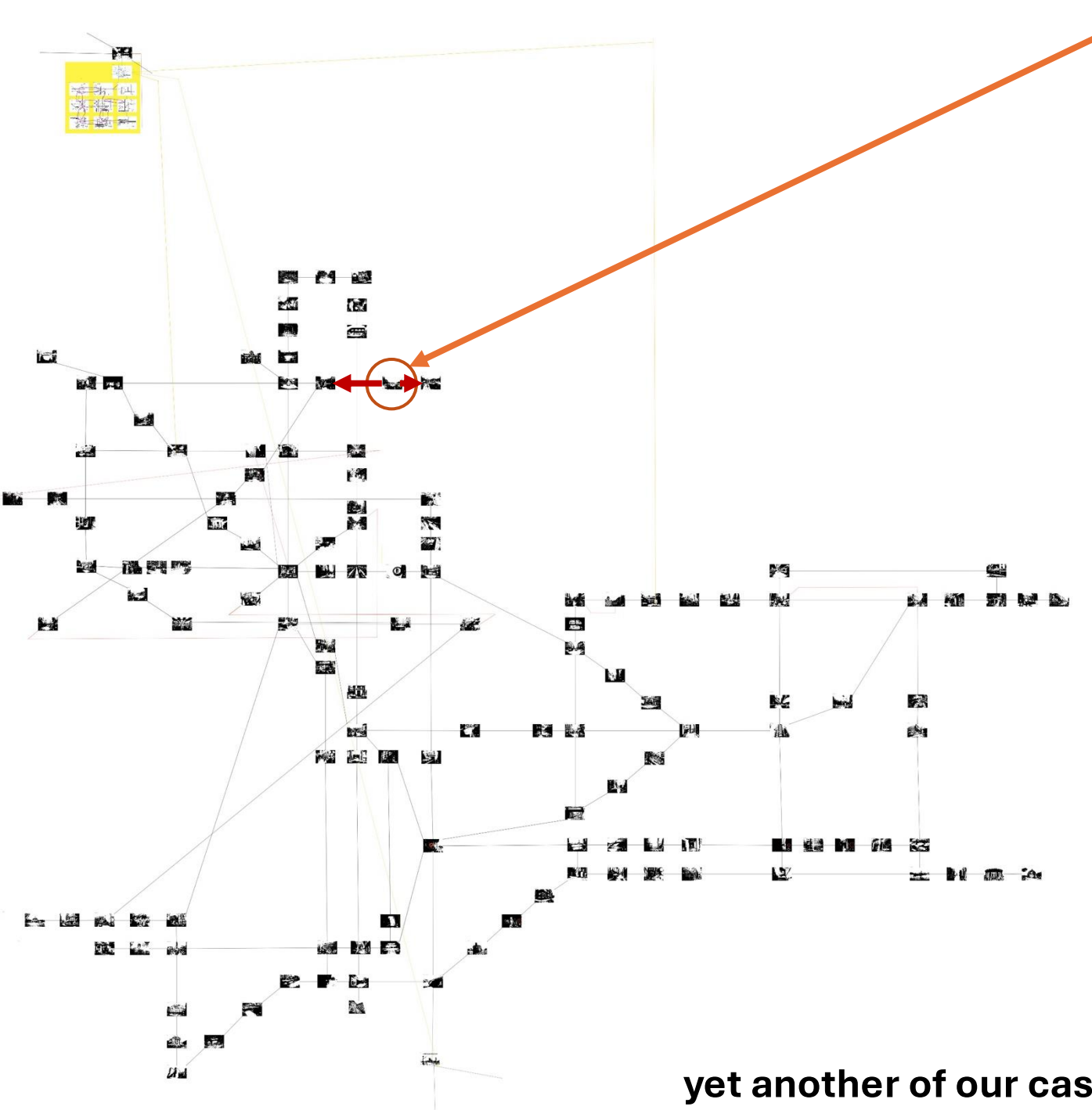
Storytelling: 4 argumentations, cardboards on table, magnifying glasses.

academic + public context



Docu & Demo

Exhibition at the Collegium Helveticum,
ETH Zurich, 15.-29.2.2024



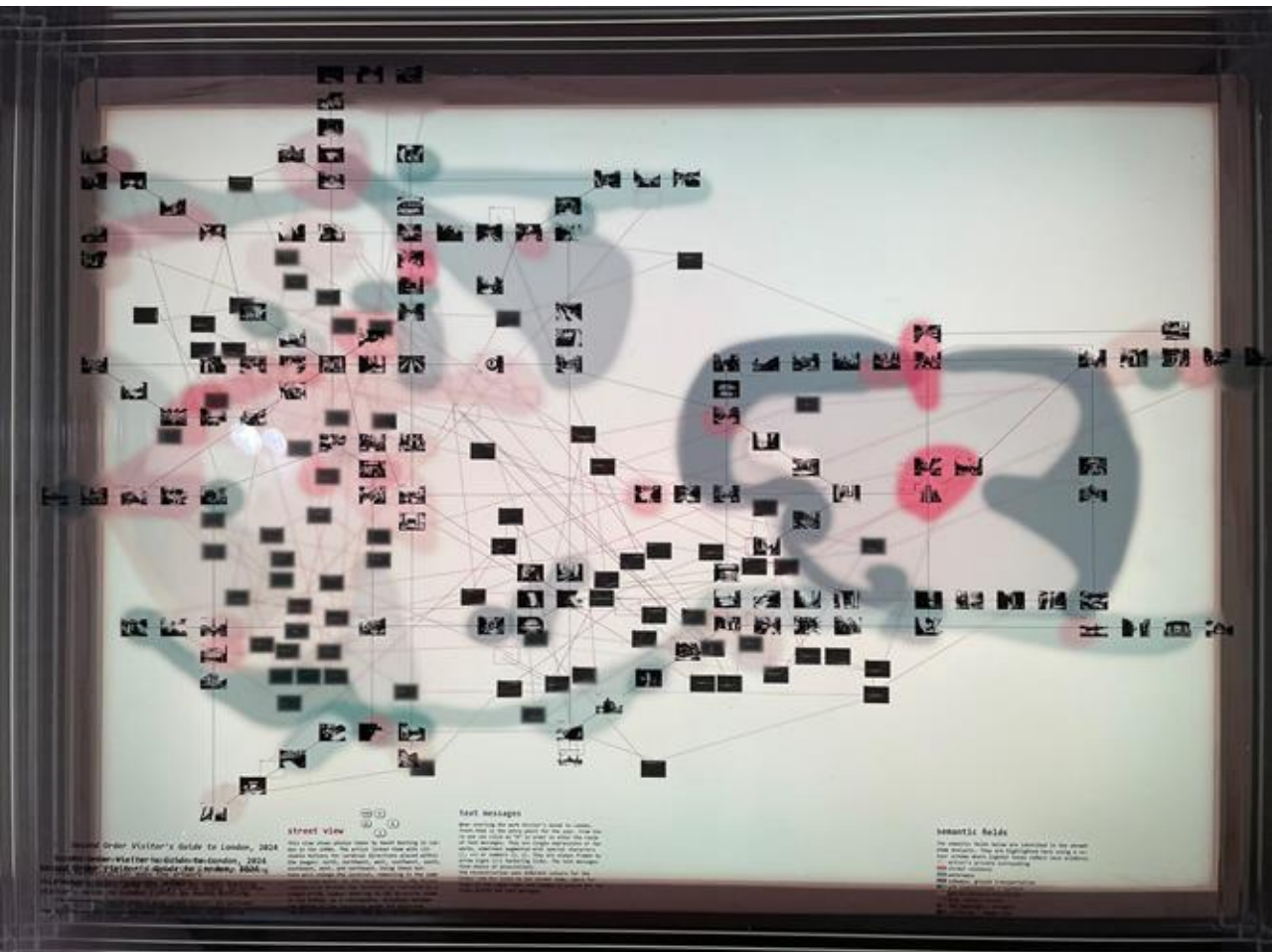
Heath Bunting: *Visitor's Guide to London*, 1995.
Interface (= detail), is a massively hyperlinked website labyrinth

COSE: *Second Order Visitor's Guide to London*, 2024.
Map layer 'street view' (= detail photos, manually arranged along cardinal directions as indicated)

yet another of our case studies ...

Analyses of a highly networked artistic website

case studies

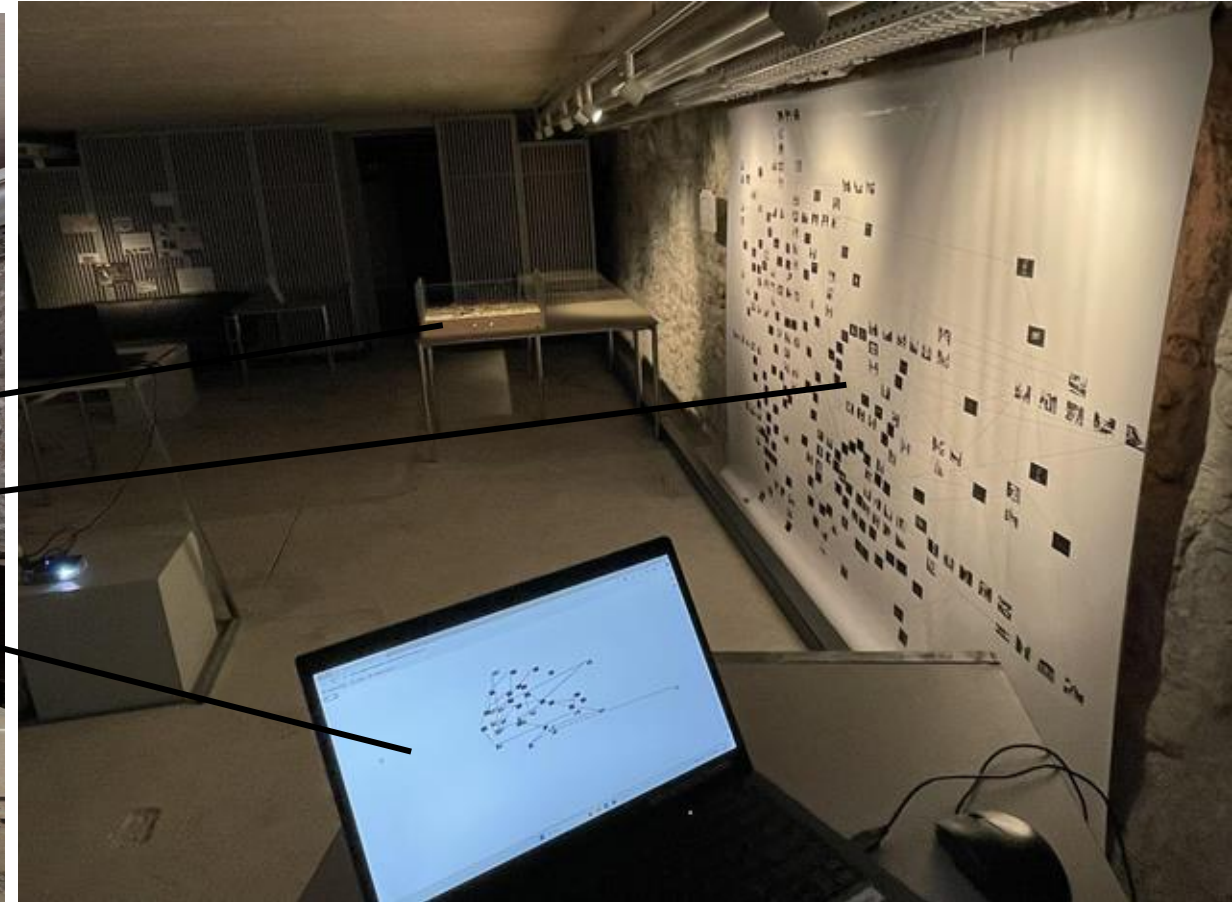


COSE: *Second Order Visitor's Guide to London* (2024)
(analysis of Heath Bunting: *Visitor's Guide to London*, 1995)

... and how it can be mapped and interpreted in various ways ...

(curated research-based) exhibitions

Docu & Demo: Archiving Programmed Media Art, Collegium Helveticum, ETH Zurich, Feb 15-29, 2024



case studies

Model set: physical 9-layer model, 3-layer poster, interactive mapping

Maps (poster, layer, dynamic) + source/reference
(COSE: *Second Order VGTL*, 2024)

Book + source/reference
(Browser Art: *Navigation*, 2022)

Argumentation (game, game guide, book) + source/reference
(COSE: *ROAMING.COM*, 2025)

3:3

(exhibitable) epistemic model complexes

Book publications (ecosystem of publications around browser art)

method/documentation
case study

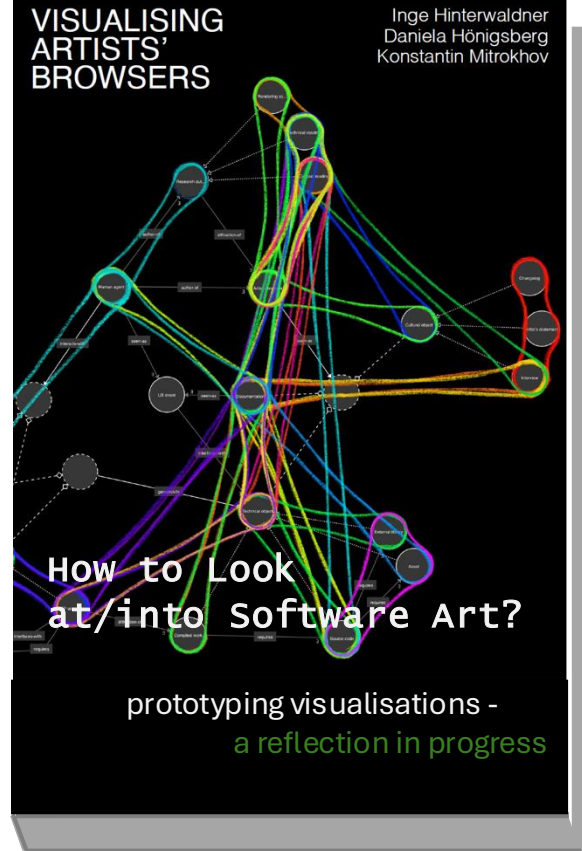
Web browsers prescribe the ways we access and navigate knowledge and communities online. Since the 1990s browser software has been an arena for artistic interventions, ranging from quirky standalone browsers to performative pieces to minimalist browser add-ons. The (im)possibility of navigation is not taken for granted and is probed, questioned, and reformulated through such software practices. We propose navigation as a method for art-historical inquiry that allows to capture multifaceted artists' browsers in action.

NAVIGATION

Reihe
Begriffe des
digitalen Bildes

eds. Inge Hinterwaldner, Daniela
Hönigsberg, Konstantin Mitrokhov,
130 p.
LMU Press: Munich 2022.

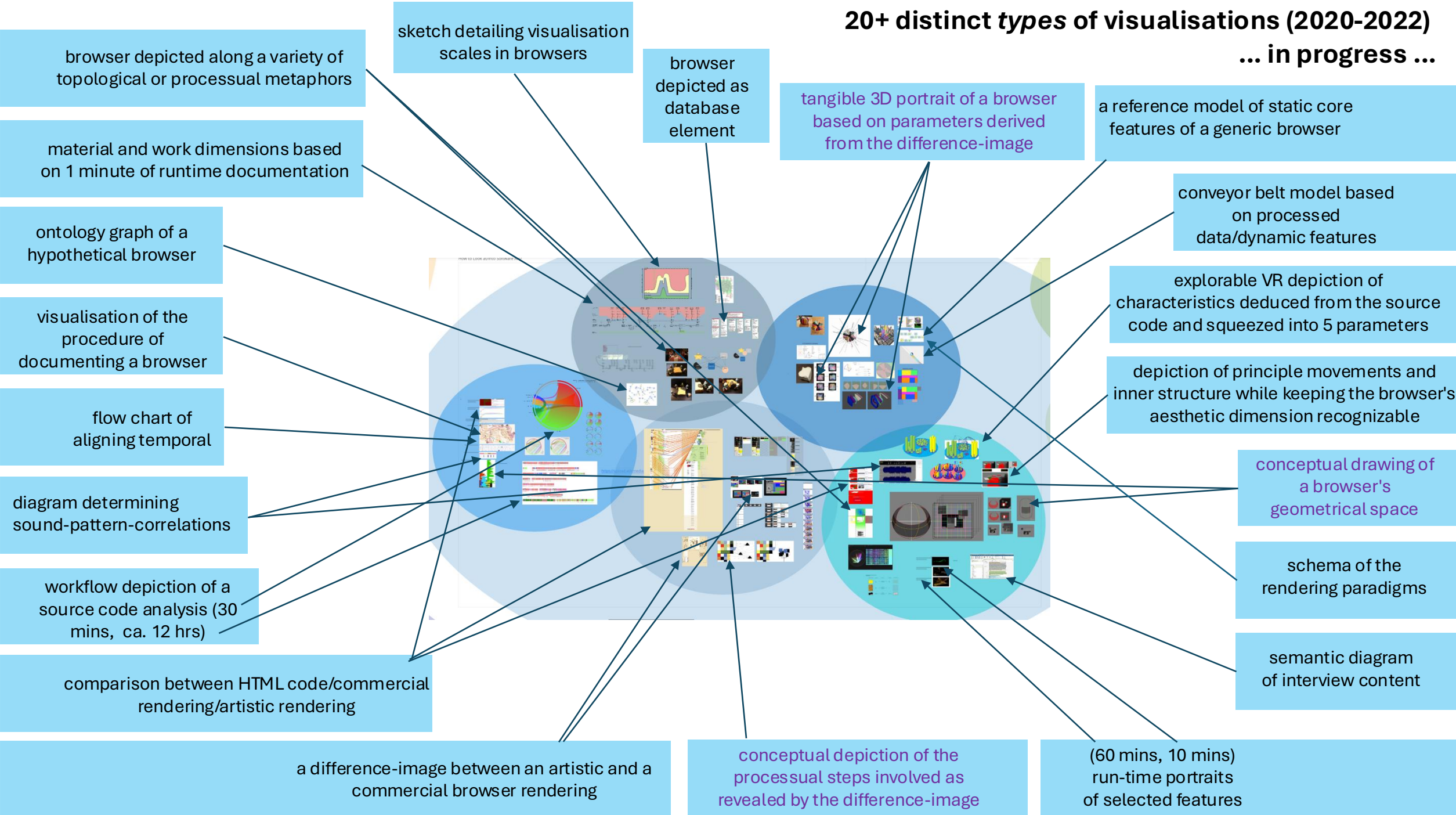
method/visualisation
survey

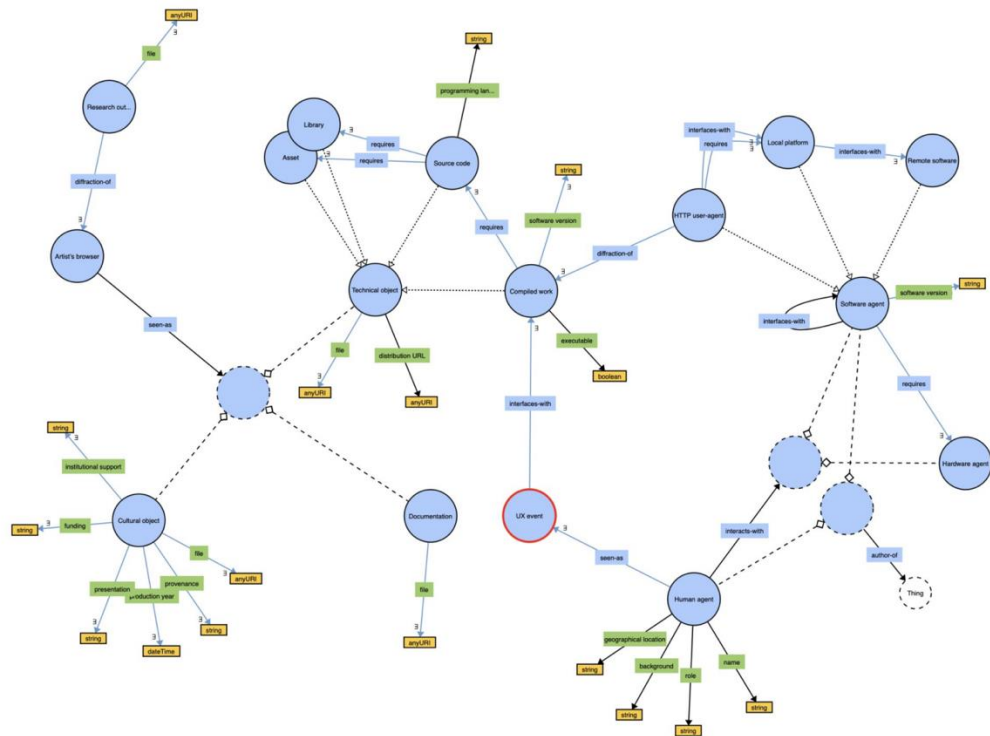


eds. Inge Hinterwaldner,
2021-2022 finished; conceptually a
work in progress, 172 p.
Intercom: Zurich

20+ distinct types of visualisations (2020-2022)

... in progress ...





SOFTWARE ONTOLOGY GRAPH

date: June/November 2021

status: finished (second iteration)

software used: WebProtégé, owl2vowl, WebVOWL (local instance)

DESCRIPTION

Context: this visualisation is part of an effort to model a software ontology of artist's browser.

Function: heuristic.

Intention: modelling a bottom-up ontology of artist's browser as a situated software class that includes multiple research perspectives.

Question: is the ontology accurate and full?

Source of data: embodied knowledge of the modelled subject.

Sequence: 1) model the ontology in WebProtégé through an iterative process with the aim of accounting for most relations, subclasses, and research perspectives; 2) export the file from WebProtégé in the format supported by WebVOWL; 3) load the file into WebVOWL running locally in a supported browser; 4) tweak the graph display parameters for better legibility with the help of "pin" tool; 5) revise the graph and make notes for the next iteration of the software ontology; 6) repeat steps 1-5 until the ontology is satisfying.

Sources & references: ontology modelling, knowledge graphs.

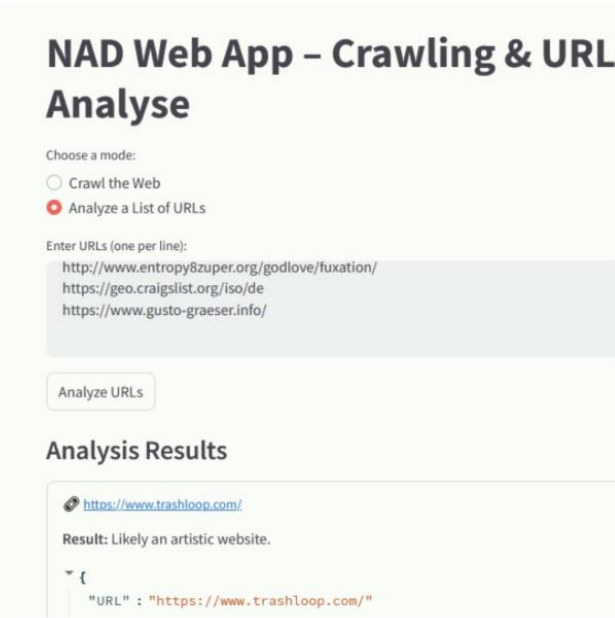
BRIEF REFLECTION

This graph is extremely useful during the process of modelling an ontology.

WebProtégé lacks built-in visualisation tools and doesn't provide a visual overview of modelled ontologies. It is difficult to have an overview of an ontology without looking at its graph that shows all the relations and annotations. When the ontology is finished, its graph could serve as a baseline vector illustration for further design work.

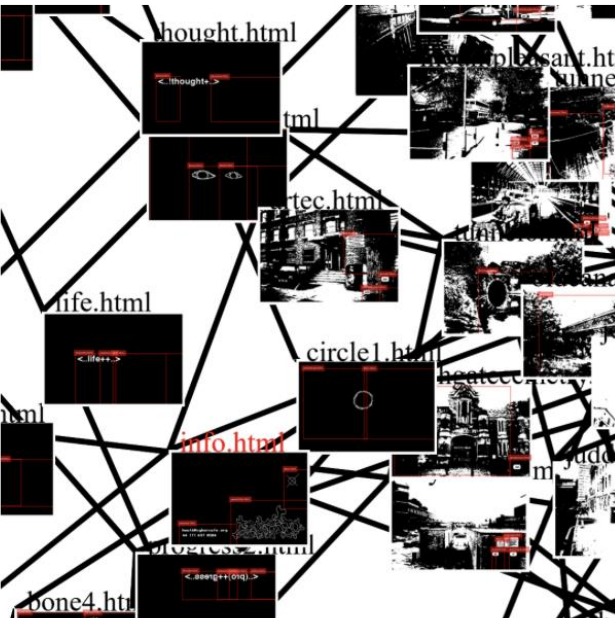
The main issue with ontology modelling is that its graph is as abstract as all other graphs.

Analytical tools



NAD | Net Art Detector Tool v0.01, 2024

The NAD Tool is a browser0based application developed by Daniela Hönigsberg to support the investigation and analysis of web-based artistic pratices and to assist researchers, curators, and students in identifying webpages that diverge from conventional design practices. [Read more ...](#)



Heat Map Web Crawler with Visual Site Mapping, 2024

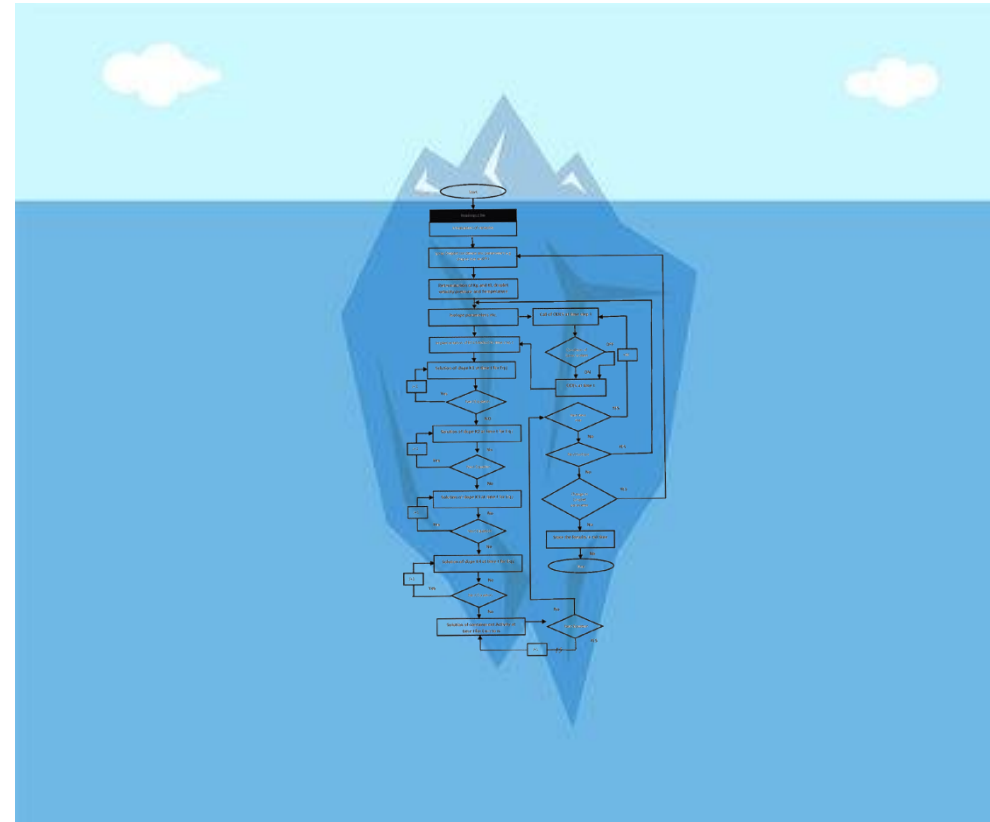
The Heatmap Web Crawler with Visual Site Mapping tool is a desktop application that crawls websites and generates visual site maps with screenshots. This software tool is built with Electron, Express, and Mermaid.js. by Emma Dickson. [Read more ...](#)

Functional portraits

(desire, not done yet on a larger scale)

Expand the levels of comparison between artworks based on an ontology

Future work: creating artworks-specific portraits of how they function



monitor output
(phenomenological level)

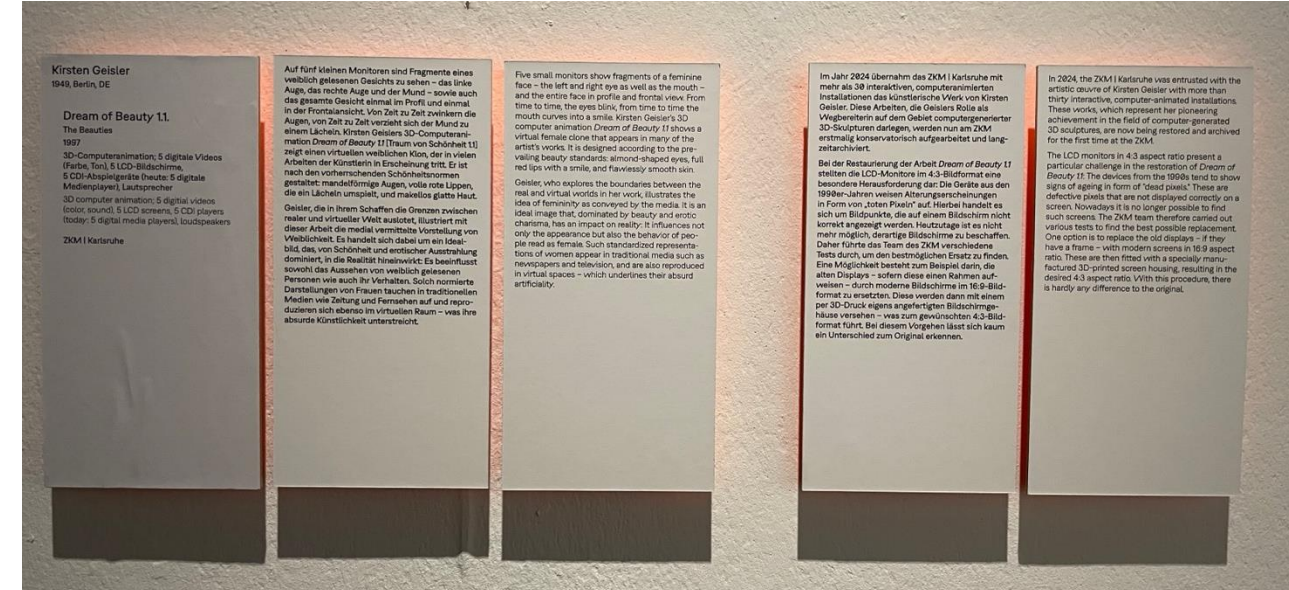
program execution
background processes
functionality

Vision: exhibitions as ready-made experimental setups 4 systematically logging the changes

- Build a community of curators / artists / technologists / conservationists interested in exhibiting online art
- Conceive exhibitions as occasions to rethink living-on-formats (online, offline, hybrid)
- Curators / artists conceptualise the next iteration
- Conceive exhibitions as occasions to log software changes
- Technologists / conservators update the works to make them ready for the show
- Technologists / conservators repair the works while on show
- They record what was done why and submit entries to a crowd source interface
- Community of curators / technicians / conservators / artists publish jointly in a quarterly (edited sum of entries)

Benefits:

- artwork-relevant record of online events that else go unnoticed (humanities' interest, history)
- solutions published beneficial for all other works suffering that were not on show at that time (artists / museums interest)
- destigmatise the fact that artworks need repair (conservators' interest, whose work becomes more visible)



ZKM exhibition "The Story that Never Ends", since 2025-2028.

Thank you for your attention!

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We are hiring!



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